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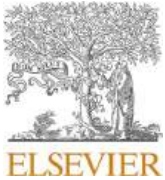


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Architecture for pavement pothole evaluation using deep learning, machine vision, and fuzzy logic

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ABSTRACT

Roads are critical for economic growth and trade but are constantly degraded by heavy traffic and adverse weather, leading to potholes that compromise safety. Traditional detection methods, like manual inspections, are labor-intensive, costly, and prone to errors. Existing automated systems also struggle with false positives, particularly in challenging conditions involving shadows, stains, or other environmental interferences. This research presents an architecture for detecting, measuring, and evaluating potholes, as well as generating maintenance recommendations. We integrate 2D images and 3D point clouds captured using the Intel RealSense D435i camera, generating a dataset of 583 images—299 containing potholes and 234 depicting environmental noise on various pavements. Each image is labeled through semantic segmentation and paired with corresponding point clouds. The architecture utilizes transfer learning with a Segformer network, achieving high detection performance with a Recall of 90.87 %, Precision of 90.01 %, Accuracy of 86.8 % F1 Score of 90.433 %, and a loss of 0.0431. The method achieves an IOU of 85.872 %, ensuring accurate diameter estimation, which contrasts with studies using lower IOU values where pothole dimensions are often underestimated due to incomplete detection. Our architecture provides reliable contour detection, facilitating the integration of image data and point clouds to estimate pothole dimensions with a depth estimation error of 5.94 mm. A fuzzy logic system processes these measurements to assess repair urgency and recommend appropriate repair techniques.

1. Introduction

Adequate road maintenance is essential for raising the economy and ensuring social connectivity. However, pavement roads face significant challenges, such as visible deterioration, manifested by potholes and cracks, leading to an increase in maintenance costs,

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which are often due to repairs being carried out at stages where the damage has already become severe. Poor pavement conditions increase the risk of accidents, reduce travel comfort, and increase the likelihood of vehicle damage, among other inconveniences. In addition, regular inspections and repairs can cause significant traffic disruptions, further affecting urban mobility and productivity. Consequently, conducting systematic assessments or monitoring the state of roads and organizing maintenance, either preventive or corrective, is crucial. These activities for pavement roads require damage evaluation based on their severity, using physical measurements such as diameter, length, and depth. Tools such as rulers, calipers, and tape measures are employed to obtain these measurements [1].

Traditionally, inspections have been laborious, costly, and subject to human error during the process, in addition to constantly jeopardizing the personnel performing them. With the advancement of technology and artificial intelligence, systems have been developed to detect damage and automatically reduce these problems. Automatic detection of structural and pavement damage has been an active area of research. According to [2], the systems proposed to address these problems can be classified into three categories: image-based systems, signal capture systems, and three-dimensional (3D) reconstruction-based systems, with image-based systems being the most popular and least expensive.

Initially, digital image processing techniques such as edge detection [3] and threshold [4] were used, assuming that damaged areas appeared darker than healthy ones. However, this assumption does not always hold since the presence of external elements such as shadows, stains, or changes in illumination leads to false-positive detection. Later, with the development of artificial intelligence algorithms such as convolutional networks or, recently, transformer networks [5], significant progress has been made in this area. Works like [6] demonstrated to be able to detect cracks in concrete structures by means of convolutional networks. Their results show that they were able to capture cracks up to 1.5 mm wide. In 2018, the same authors [7] added to their model the detection of corrosion in metal structures, which is useful for observing the health of a structure that combines concrete and metal. While previous works have focused on crack detection in buildings and infrastructure [6], others have addressed damage assessment [8] in road pavements, including cracks and potholes.

Recent studies, such as [9,10], have demonstrated significant advances in pothole detection using semantic segmentation techniques based on computer vision and deep learning, achieving accuracies of over 90 % under ideal conditions. However, these systems face critical challenges: their sensitivity to light variations during image capture, especially in environments with dynamic shadows or intense reflections, can degrade their performance, while visual artifacts such as stains, manhole covers, repair patches or heterogeneous pavement textures generate false detections when mistaken for pothole patterns. Additionally, reliance on large volumes of manually annotated data at the pixel level increases costs and development time.

Signal capture systems employing vibration sensors and accelerometers [11,12], which are typically found in smartphones, are investigated for pothole identification. These systems record vibrations and acceleration events when a vehicle crosses a pothole, analyzing the data with multilayer neural networks to identify road damage. Although cost-effective and straightforward to implement, these methods have notable limitations. Detection accuracy can be affected by vehicle speed, suspension type, and road conditions, leading to a high incidence of false positives and negatives. In addition, variability in the placement and attachment of devices to vehicles can result in inconsistencies in the data collected. These challenges limit the effectiveness and reliability of vibration sensor and accelerometer-based methods for accurate road bump assessment.

The approaches described above have limitations with respect to the inability to capture spatial information such as length, width, or depth. Considering these facts, devices that allow 3D reconstruction have been used, such as lasers, stereo cameras, or even monocular cameras [13]. Different industries widely use these devices, which allow them to collect spatial data with high precision [14–16]. In addition, previous studies have utilized these tools to identify road deterioration, including potholes and cracks [17,18]. However, devices that offer an extensive range, precision, and resolution often come with a substantial price tag, occasionally exceeding twenty thousand dollars. The Lidar sensor generates pulses that calculate the distance between the sensor and the surface by measuring the time it takes for the pulses to reflect back. In contrast to lithium technology, depth cameras operate using one or two digital cameras to take two-dimensional images and infer the depth of objects within the scene through stereoscopic [19] or structured light [20] methods. Depth cameras are less expensive than lidar cameras, although they sacrifice accuracy and range. However, these systems balance cost against the precision of depth estimation. At present, the speed of the vehicle carrying the capture devices leads to an average depth error ranging from one centimeter (cm) to three cm and, in some cases, exceeding six cm. Furthermore, existing research has concentrated mainly on damage detection and classification [21–23], often neglecting the essential task of assessing pavement damage to quantify damage. This quantitative assessment is crucial as it provides objective metrics for determining the severity of the damage, estimating repair costs, and prioritizing maintenance interventions based on the actual condition of the pavement rather than just its visual classification. While other works, such as [13] and [24], perform measurement estimation from a single data source, their work is only performed in a static position and requires several images from different views of the pothole. This allows greater control over ambient noise and damage selection but is inadequate to improve the speed and evaluation of a large road in a short time. We emphasize that none of these works recommend rehabilitation treatments.

The main contribution of this work lies in providing a comprehensive evaluation architecture that not only detects and measures potholes but also assesses the urgency of repair and recommends the most suitable repair technique, facilitating informed decision-making for road maintenance. To the best of our knowledge, this integrated approach for pothole assessment and repair recommendation has not been previously reported in the literature. The presented approach focuses on low-cost data integration, specifically combining 2D and 3D data sources. This combination not only improves damage assessment but also enables repair recommendations, which is a key advance over existing methods focused solely on detection with separate 2D or 3D data. First, the architecture performs a semantic segmentation to identify the damaged areas in RGB images, and then the coordinates of the pothole contour are stored. Subsequently, the architecture obtained the depth and diameter of the pothole using the point cloud, focusing only on the damaged

areas. Following this, the architecture applies a fuzzy logic approach to evaluate the severity of the pothole and determine the urgency of repair. This architecture uses depth and diameter measurements as input parameters to generate an assessment of the damage and provide a recommendation for the most appropriate repair technique.

This study addresses the challenges associated with the limitations of traditional and contemporary pavement damage detection and assessment methods. The limitations include the detection of false positives in 2D images, the high cost associated with conducting inspections using specialized devices, and the inherent risk of manual inspections, the presence of external elements such as shadows, stains. To overcome the drawbacks of image-based methodologies when detecting potholes, we apply a transfer learning approach focused on feature extraction by implementing a Transformer-type network known as Segformer [25] since it has attention modules focused on both local and general features of the objects to be detected. We constructed a data set with 583 images of potholes on flexible and concrete pavements, including their point cloud on mixed ambient situations collected with an Intel D435i depth camera. We trained the Segformer network with this new dataset to extract the characteristics that allow us to enhance pothole detection. We tested the architecture in an urban environment during sunny days at a temperature of approximately 30 degrees Celsius, at speeds between 20 and 50 kilometers per hour, with the capture device installed on a vehicle at the height of 1.25 m from the surface, ensuring that the capture is parallel to the surface.

We evaluated the impact of potholes by creating a fuzzy logic system that simulates the decision-making process of inspectors following the severity classification guide. [26] The manual outlines the process for assessing the extent of damage to the pavement's surface and structure, using the diameter and depth of each pothole as input parameters. The manual categorizes the severity into low, medium, and high levels. In contrast, we suggest using five categories: minimal, low, medium, moderate, and high to more precisely classify the physical dimensions (diameter and depth) of each variable. The architecture produces two outputs intended for both practicality and efficiency. The initial output indicates how urgent/critical repairs are needed to keep the road in optimal condition, considering the level of damage observed, and classifies the urgency into categories: no urgency, low urgency, medium urgency, and high urgency. The second output provides a technique recommendation for damage repair for each pothole evaluated: 1) routine preservation, 2) crack sealing, 3) crack felling, 4) patching, 5) seal irrigation, and 6) reconstruction.

We summarize the paper's contributions as follows:

- This paper aims to contribute to generating an automated inspection considering pothole damage severity in urban streets and provide a fuzzy decision support system that recommends the maintenance technique that repairs the damage for each pothole detected.
- We construct a new dataset that integrates 583 images, considering 299 potholes and 234 images with environmental noise (for example, shadows, stains, and lighting changes) from rigid and flexible pavement. We generate semantic labeling for the potholes and integrate their point clouds. Furthermore, only for 11 potholes, we manually measured the actual depth and diameter to determine their severity level. Data collection was carried out with an Intel D435i structured light depth camera, available at: <https://github.com/Garaymc54/Pothole-DCI>.
- A transfer learning was implemented in a Segformer network as an inference model trained to segment potholes. The transfer learning process utilized 583 images of potholes and non-potholes for feature extraction. Of these 583 images, 404 were assigned to the training set, 119 to the validation set, and 60 to the test set. In particular, only the training set was subjected to data enhancement techniques, where each image was transformed into three by applying 180° rotation and mirror flipping, increasing the number of images in this set to 1212. The results achieved were 86.8 % accuracy and 85.872 % for the mean IoU to detect potholes. A higher IoU indicates more accurate detection of the pothole contours, which directly contributes to more precise measurements of pothole dimensions such as diameter and area. These results demonstrated competitive performance compared to the state of the art; see 2.
- Our architecture is designed to facilitate the quantification of the potholes. It integrates the preprocessing of images and cloud point data to detect and semantically segment potholes. Using data integration and transfer learning with the Segformer, we have achieved an average error of 7.82 mm in depth estimation. Data integration allows us to focus only on the damaged area, reducing the amount of data to be processed.
- The fuzzy logic system developed in this work uses the diameter and depth of the pothole as input to generate two key outputs. First, the severity or urgency level of repair helps prioritize interventions based on the risk the pothole poses to road safety and pavement integrity. Second, considering the individual characteristics of the pothole, the system provides a specific recommendation for the most appropriate repair technique to apply to the pothole.

The rest of the sections are organized as follows. Section 2 State of the Art analyzes previous work. Section 3 Materials and Methods describes the functionality of the architecture in detail. Section 4 Results presents the results obtained. Section 5 Discussion describes the analysis of the results. Section 6 Conclusion mentions the conclusions derived from this work.

2. State of art

Pothole detection on roads is crucial for maintaining road infrastructure and transportation safety. Technological evolution has enabled the development of multiple approaches based on signal analysis techniques, digital image processing, or 3D reconstruction. This section reviews works from each approach, highlighting their strengths and limitations. Table 1 compares recent work, showing characteristics of the hardware used to collect images, detection techniques, and whether they estimate depth and propose maintenance recommendations.

2.1. Signal analysis

The use of mobile devices, specifically their accelerometer, gyroscope, and, in some cases, the microphone, has been highlighted in research such as [31]. The principle involves installing a smartphone in a vehicle and ensuring that it passes over road damage to collect data from the internal sensors. Some studies, such as [32] and [33], use machine learning to detect damage, achieving accuracy rates above 90 %. Other studies, such as [31], use the visual representation of signals from the accelerometer and gyroscope, processing them with a convolutional network. However, a limitation of this type of system is the need to pass over road damages to detect them, subject to variables such as the condition of the car's suspension.

2.2. Road damage detection in 2D

Damage detection in 2D (two dimensions) has been used to determine the deterioration zone and classify the type of damage present on the road. In this sense, these works are not oriented to quantify the kind of damage but to locate it, generating their coordinates and displaying them.

J. et al. [34] presented a deep learning-based model that uses SqueezeNet, U-Net, and Mobilenet-SSD to detect, classify, and assess the severity of road cracks. The first network, SqueezeNet, distinguishes between linear damages (cracks) and area damages (potholes). The second network, U-Net, segments the damages, and the third network, Mobilenet-SSD, classifies the specific type of damage. The damages identified include longitudinal cracks, transverse cracks, alligator cracks, patches, and potholes. The assessment evaluates the damage based on the percentage of damage in the image without considering specific dimensions such as length and depth.

R. et al. [9] introduced an approach for road pothole detection using semantic segmentation algorithms and stereoscopic images. The method generates a new image using stereo images, allowing for a more apparent distinction between damaged and undamaged road areas. Although this technique improves the visualization of damage, it does not perform a three-dimensional reconstruction of the area or estimate the severity of the damage. The main contribution is the network that generates these enhanced images.

2.3. Damage reconstruction in 3D

The following section presents several studies that address the detection and quantification of road deterioration, with a primary focus on pothole identification. A key feature of all these works is the use of 3D data or the three-dimensional reconstruction of the area to determine the depth of the damage. The different approaches and techniques used in the segmentation and classification of these deteriorations are described, highlighting both the methodologies employed and the limitations of each study.

M. R. et al. [30] proposed a methodology to process data from the Riegl VQ-450 mobile laser scanner to identify and quantify road deterioration, such as potholes and bulges. The methodology includes editing point clouds, determining roughness, segmenting deteriorations, and calculating geometric parameters to classify severity. To detect damages, the authors used the point cloud and an algorithm that classified a point as part of the deterioration if its depth was greater than 13 millimeters; with this information, they created binary images to segment the damage. The results show accurate segmentation and classification, improving safety and efficiency by not affecting traffic during data acquisition. However, the devices used are costly and increase maintenance costs.

Bosurgi et al. [17] used the HiProSS (High Performance Road Survey System) developed at the University of Messina to capture three-dimensional data from road surfaces. This device combines 3D cameras and linear lasers to obtain images with 3D information of the surfaces using optical triangulation techniques. For the semantic segmentation of potholes, they employed a semantic filtering process that separates information related to potholes from other defects, such as cracks. The segmentation is based on the intensity scale of the image, which, in the three-dimensional representation, corresponds to the different heights of the pavement. The algorithm

Table 1
Comparison of the State of the art works.

Article	Approach	Hardware	Data type	Data Availability	Depth estimate	Task
G. et al. [17]	Digital image processing	HiProSS	Point Cloud	No	Yes	Detection
H. et al. [18]	Deep learning DCNN	Laser SICK LMS 511	Point Cloud	No	Yes	Detection
R. et al. [9]	Disparity maps	ZED Camera	Stereo images	Yes	No	Detection
A. et al. [27]	Fotogrametry	Sensor Laser Camara digital	Point Cloud	No	Yes	Detection
Peralta-Lopez, et al. [28]	Deep learning CNN	ZED Camara	Image 2D	No	No	Detection
I. et al. [29]	Deep learning Transformer	ZED Camara	Image 2D	No	No	Detection
M. R. et al. [30]	Point cloud segmentation	2 Laser iegl VW-450	Point Cloud	No	Yes	Detection Evaluation
R. et al. [13]	Deep Learning CNN	3D GO SCAN	Images Disparity maps	Yes	Yes	Detection
This work (2024)	Deep Learning Data integration	Real Sense D435if	Image 2D Point Cloud Label	Yes	Yes	Detection Evaluation Recommendation

identifies areas of lower altitude, excluding the background and other elements not related to the potholes.

The above studies demonstrated that it is feasible to assess road damages; however, the devices used are costly, increasing maintenance costs. Additionally, due to the large size of the 3D data collected, a large storage capacity and high computational power are necessary.

A. et al. [27] presented a cost-effective technique for 3D reconstruction of road potholes using a system composed of a camera and a laser pointer. This technique combines a Structure-from-Motion (SfM) algorithm with laser triangulation to generate 3D point clouds and detect potholes. The authors used the RANSAC algorithm to segment the potholes in the point cloud directly. Although this is a low-cost technique, changes in lighting or contrast can sometimes make the laser invisible in the images; however, the system is still able to obtain accurate depth data. The disadvantage of this approach is that outliers can produce errors when calculating depth.

H. et al. [18] presented a method that integrated images and point clouds to detect potholes and measure their depth. The data is collected using a panoramic camera installed on top of a car and a SICK LMS 511 laser sensor mounted at the rear. The vehicle travels at an average speed of 12 km/h. The point cloud density generated at this speed is approximately 130–190 points/m². Images and the DeepLabV3 library with the Xception-65 convolutional network are used to detect the potholes. The depth is determined by measuring the distance between the pavement and the points belonging to the pothole. The average distance is calculated, and if it exceeds a predetermined threshold, it is classified as a pothole; otherwise, it is classified as a patch.

In 2024, R. et al. [13] proposed a method for real-time pothole segmentation, volume, and depth estimation using a monocular camera combined with the 3DPredictNet deep learning model. This approach represents a significant advancement, as it enables the estimation of depth data solely from monocular camera input. However, this method has a notable limitation: the data must be captured in a static manner, requiring the camera to circle the pothole of interest from different angles. This constraint results in longer data acquisition times compared to the methodology proposed in our work.

Upon analysis of Table 1, it is evident that only a few of the previous studies providing open-access datasets include both images and their segmentation labels. In contrast, this work offers a more comprehensive dataset, which includes not only 2D images and their corresponding pothole segmentation labels but also point clouds associated with the images, along with detailed physical data on the diameter and depth of 11 potholes. Furthermore, while most studies focus on damage detection, some even using 3D information, none of them address the evaluation of the urgency of repair or offer specific recommendations for pothole repair. This differentiating aspect is key in our proposal, as it not only detects potholes, but also assesses the urgency of their repair and provides recommendations for the appropriate treatment, representing a significant contribution beyond traditional approaches.

3. Materials and methods

Fig. 1 illustrates the architecture proposed for the detection, segmentation, evaluation, and generation of maintenance recommendations for road potholes. The process begins with data collection using a vehicle equipped with sensors to capture images and point clouds. These data are extracted and stored for subsequent preprocessing, where the images and point clouds are filtered and resized. The following steps include pothole detection, parameter estimations such as diameter and depth, and finally, an evaluation that determines the urgency/severity of the pothole and recommends the appropriate repair technique.

3.1. Data Collection

Fig. 2 shows how the depth and diameter of potholes are measured according to the Distress Identification Manual for the Long-Term Pavement Performance Program by the FHWA [35]. The figure illustrates a representation of the road with a 100 mm pothole and a cross-sectional view of the road to show how the depth is measured. The depth measurement is taken from the pavement surface plane to the deepest point of the pothole, while the diameter is measured between the farthest points of the pothole perimeter.

Hence, the problem to solve in this paper is how we automatically detect potholes, determine the quantification of their severity, and provide how urgent it is to repair them. In addition, we highlight that with the quantification of the potholes, we can recommend the repair maintenance technique.

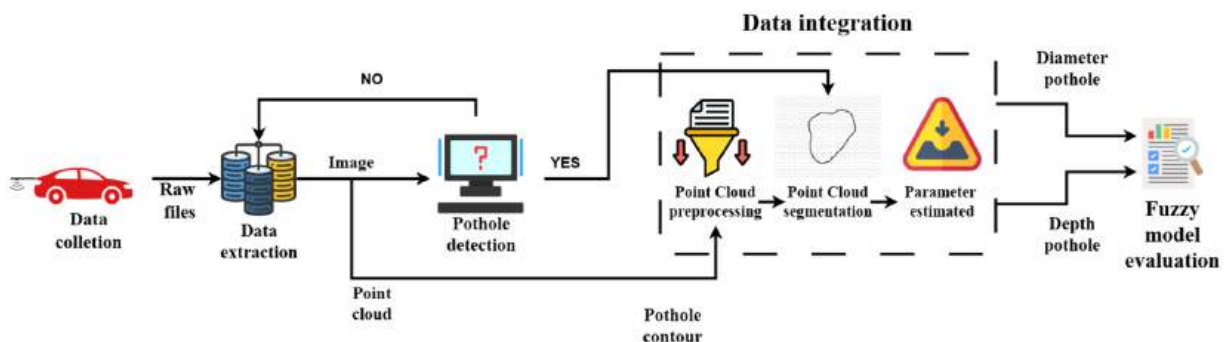


Fig. 1. Architecture for pavement pothole evaluation and generate maintenance recommendation.

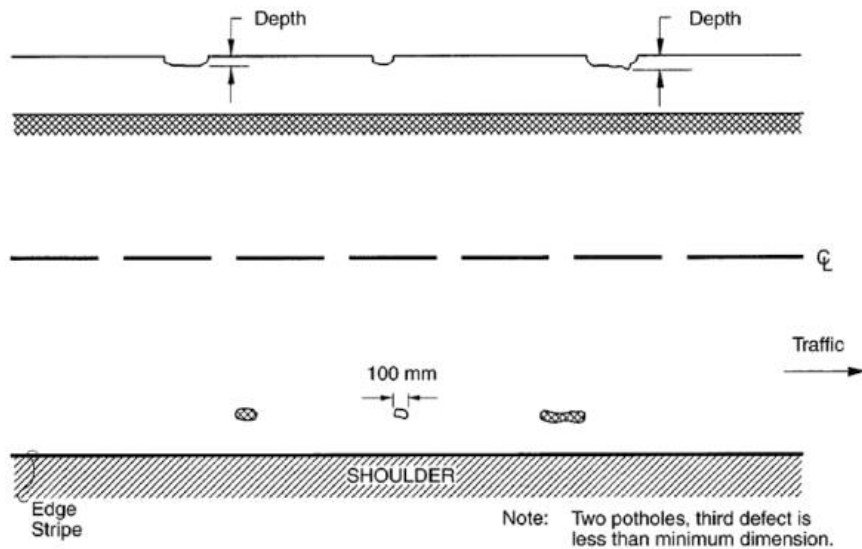


Fig. 2. Measurement of potholes by FHWA [35].

In this sense, for the first step of our architecture, we use a RealSense D435i sensor for data collection, image capture, and point clouds to provide essential raw data for analysis. This device is equipped with a stereo camera and an infrared sensor. It was mounted on a vehicle using a tripod with suction cups to ensure stability, as shown in Fig. 3. The distance between the ground and the RealSense D435i sensor is 1.25 m. The vehicle traveled between 10 and 50 kilometers per hour in an urban environment. During the journey, the sensor captured images and pointed clouds at the pavement surface, allowing for a three-dimensional scene representation. Different sensor elevations were experimented with, spanning from 1 m to 1.5 m with increments of 5 cm. The preferred height of 1.25 m yielded the most precise depth estimation, aligning closely with the actual measurements.

The Intel RealSense D435i was selected as it offers an optimal balance between cost and performance. While high-end LiDAR sensors can cost upwards of \$20,000, the D435i provides reliable depth sensing capabilities at approximately \$500, making it a cost-effective solution for widespread implementation in road monitoring systems. Also, it can detect depths from 0.105 m to 10 m, which is particularly suitable for our application, as we mounted the sensor at 1.25 m from the ground. This working range allows for accurate detection of both shallow and deep potholes, capturing defects as small as 2.5 centimeters in depth. The sensor's built-in IMU (Inertial Measurement Unit) helps compensate for vehicle movement, during data collection, improving the accuracy of depth measurements even at varying speeds (10–50 km/h). Additionally, its ability to operate in outdoor conditions and its resistance to infrared interference from sunlight makes it particularly suitable for road surface monitoring. Fig. 4.

The sensor specifications during collection are as follows:



Fig. 3. Vehicle sensor installation.

- Operating range (minimum-maximum): 0.3–3 m.
- Depth field of view: 87° x 58°.
- Minimum depth distance at maximum resolution: 28 cm.
- Depth accuracy: less than 2 % at a height of 2 m.
- RGB frame resolution: 480 × 848.
- RGB frame rate: 30fps.
- RGB sensor resolution: 2 MP.

Although the sensor has the capability to capture images at a higher resolution, using this setting limits the frame rate to 15 fps. This results in the omission of pavement surface during data collection, affecting the continuity and accuracy of the captured data.

Data collection was conducted on a tertiary road during the hours of 9 am to 12 pm on sunny days to address the lighting conditions that are present most of the time in the city of Culiacán, Sinaloa, Mexico. The selected routes included rigid and flexible pavement roads to ensure a diversity of surface types. In addition, data were collected under moderate traffic conditions, where vehicles were present but did not obstruct the data acquisition process.

After collection, the architecture stores the captured data in a *.bag* file, acting as a video sequence and allowing the extraction of each sensor's recording frame. On average, every 5 s of collection requires 500 MB of storage. A Python script is used for data extraction using the *pyrealsense2* library. This library facilitates the extraction and storage of RGB images, depth images, and point clouds from each frame, ensuring that they are linked. Each type of file is saved in a different folder: one for the images, another for the point clouds, and another for the depth images. These three files share the same name but have different extensions, allowing them to be associated with each other.

3.2. Dataset extraction

For the architecture, it is crucial to establish a careful correlation between the images and the point clouds to perform adequately. Furthermore, the elements of the data set must represent the characteristics of flexible and rigid pavements in the presence of potholes and external noise, such as shadows, stains, or changes in lighting. This diversity of context strengthens the database and enhances the architecture's adaptability when faced with new images to determine if there is a pothole.

Images were chosen under different conditions to enhance the model's robustness and enable it to distinguish between various scenarios. The images selected for the data set included potholes with and without shadows, variable lighting conditions, odd elements on the road, and other types of noise, as well as healthy pavement with and without noise. A total of 583 images were selected to compose the dataset; 284 were labeled as not containing potholes, and 299 were labeled as potholes, providing a balanced and diverse set for practical training. The labeling process is described below.

3.2.1. Image preprocessing

The preprocessing carried out on each image included in the dataset has focused on four key points: 1) grayscale conversion, 2) resizing the images, 3) semantic labeling, and 4) data augmentation. We explain each preprocessing stage for the image analysis in the following subsections.

Grayscale conversion is performed to unify the color information into a single intensity channel, simplifying the data structure, and reducing the complexity of the information that will be processed. This process streamlines the color information in the image by transforming the values from the red, green, and blue channels (RGB) into a single grayscale scale that represents the perceived luminance or brightness, ranging from black to white. For each pixel in an image, the luminance Y is calculated using a formula that assigns specific weights to each RGB channel, reflecting the differential sensitivity of the human eye to primary colors. The commonly used equation for this calculation is the following.

$$Y = 0.299R + 0.587G + 0.114B \quad (1)$$

R , G , and B represent the values of the red, green, and blue channels of the pixel, respectively. In addition to converting the image to grayscale, the process includes transforming pixel values into a tensor, a multidimensional mathematical entity essential for data handling in deep learning and neural networks.

Deep learning networks are trained to function with specific characteristics, one of which is the size of input data. The Segformer model network implemented in this work requires a consistent dimension of 512x512 pixels. This is because the input layers are designed to process information at this exact size. If this requirement is not satisfied, the network will not be able to start the training process.

As part of the architecture, we carry out a transfer learning approach implementing a Segformer transformer. Hence, we need to perform semantic labeling to provide the Segformer with the examples required to perform the feature extraction and recalibrate the



Fig. 4. Graphical representation of data collection.

network. Semantic labeling delineates the object of interest within the image. In this sense, this process involves defining the perimeter of the potholes and generating a file that indicates the pixel coordinates that belong to this perimeter. These coordinates are used as input along with the image for Segformer, allowing it to identify and focus on the specific part of the image where the pothole is located, as shown in Fig. 5. This labeling was performed using the Roboflow web application [36].

We apply data augmentation to the training dataset to reduce the operative time to label new examples, prevent overfitting, and improve the Segformer accuracy. We perform geometric transformations to simulate changes in the appearance of the real-world and obtain various scenarios. Specifically, the data augmentation process is divided into two tasks: 1) 180° image rotation and 2) mirror image flipping. We applied data augmentation techniques only to the training dataset (404 images), which represents 70 % of the original dataset (583). Hence, the training set is generated by 1212 images. Examples of data augmentation generated are shown in Fig. 6.

3.3. Pothole detection

In this section, we explain how we determine that there is a pothole in an image by applying a transfer learning approach using the Segformer transformer [25]. The Segformer performs a semantic segmentation to identify the object of interest. We train the Segformer with our dataset, which contains images of potholes provided after the data augmentation. Once the pothole is segmented, we extract the cloud point data only for this area, helping to determine the depth and diameter in the data integration stage, which is described in Section 3.4.

3.3.1. Transfer learning for pothole detection

We selected SegFormer as the primary segmentation model due to its accuracy and computational efficiency balance, according to the analysis presented in [25]. As shown in Fig. 7, SegFormer-B2 achieves a mean Intersection over Union (mIoU) of 46.5 %, outperforming DeepLabV3 + /R101 (44.1 %) with significantly fewer parameters (27.5 M vs. 62.7 M) and lower FLOPs (62.4 G vs. 255.16 G). Moreover, SegFormer offers faster inference speeds (24.5 FPS).

Transfer learning is crucial in this work. Transfer learning is a machine learning technique in which a model developed for one task is reused as a starting point for another similar task. In our case, SegFormer, pre-trained on ImageNet, was leveraged to tackle the task of road pothole segmentation. This method provides significant advantages, as the model has already learned to segment and process a broad range of visual features from its initial ImageNet training.

Segformer includes an encoder and decoder with attention modules that focus on local and global features of the detected areas. SegFormer is known for its efficiency in semantic segmentation tasks. Initially, the network was pre-trained on the ImageNet dataset, consisting of 1000 categories and 1.2 million images, which allowed it to learn and identify general visual features.

In this work, we did not fine-tune the model's architecture since SegFormer has been inherently designed for semantic segmentation, aligning well with the task of road pothole segmentation. Hence, we first divide the original dataset of 583 images into three subsets: 1) training 70 % obtaining 404, 2) validation 20 % obtaining 119, and 3) testing 10 % obtaining 60. Then, we apply only data augmentation to the training dataset, obtaining 1212 images. Therefore, we focused on retraining the model's weights using our training dataset. We constructed the dataset to enhance the model's ability to accurately recognize and segment road potholes, consequently adapting the pre-trained model to our specialized context. By re-training the model's weights, the general knowledge acquired from the ImageNet dataset was transferred to our specific application, enabling the model to recognize the unique features of



Fig. 5. Semantic labeling example.

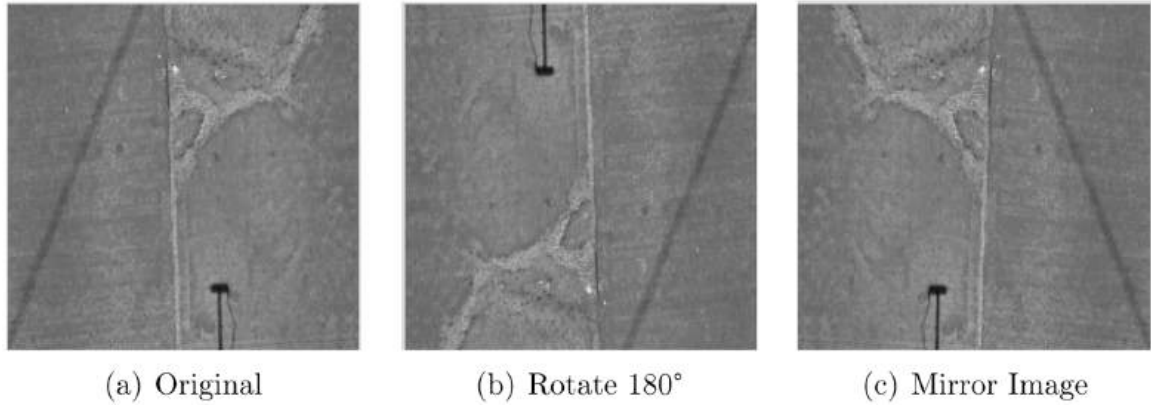


Fig. 6. Data augmentation examples.

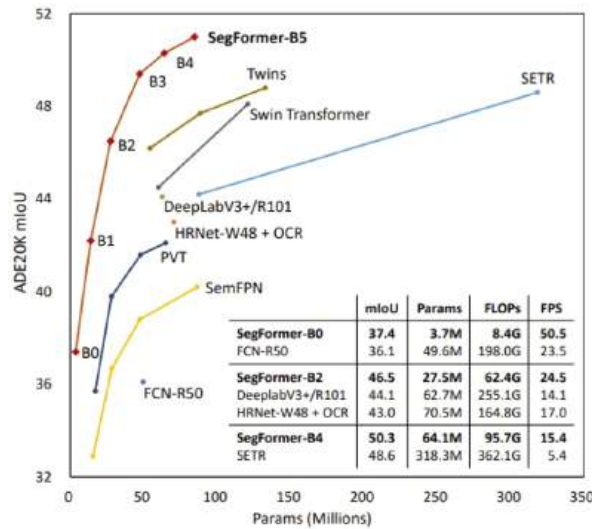


Fig. 7. Segformer comparison with others semantic segmentation models [25].

potholes in various environments. Fig. 8.

3.3.2. Training

The SegFormer model was re-trained employing a transfer learning approach, using the training dataset of 1212 images. The training was configured to run 1000 epochs with an early stop configuration, a learning rate of 0.001, and a batch size of 8 images. The cross-entropy loss function was used to optimize the model's performance. The training process was monitored using the validation loss metric (val_loss), with early stopping applied to prevent overfitting.

3.3.3. Validation and testing setup

The testing and validation are performed with their respective sub-datasets; both contain images not seen during the training phase. Due to the lack of publicly available datasets containing both images and corresponding point cloud data with manual pothole measurements, we were unable to validate the architecture on independent test sets. The validation set is used after the training phase has been completed to objectively evaluate the model. Then, the testing dataset allows us to determine the final model performance over unseen examples. The metrics used in this research are Recall, Precision, F1 Score, Mean IoU, and Validation Loss; each one is described below.

Recall measures the model's ability to correctly identify potholes present in the pavement by calculating the proportion of true positives over the total number of real potholes. A high recall value indicates a high detection rate, meaning that the model is effective in not missing potholes in the images. The recall equation is presented in (2).

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

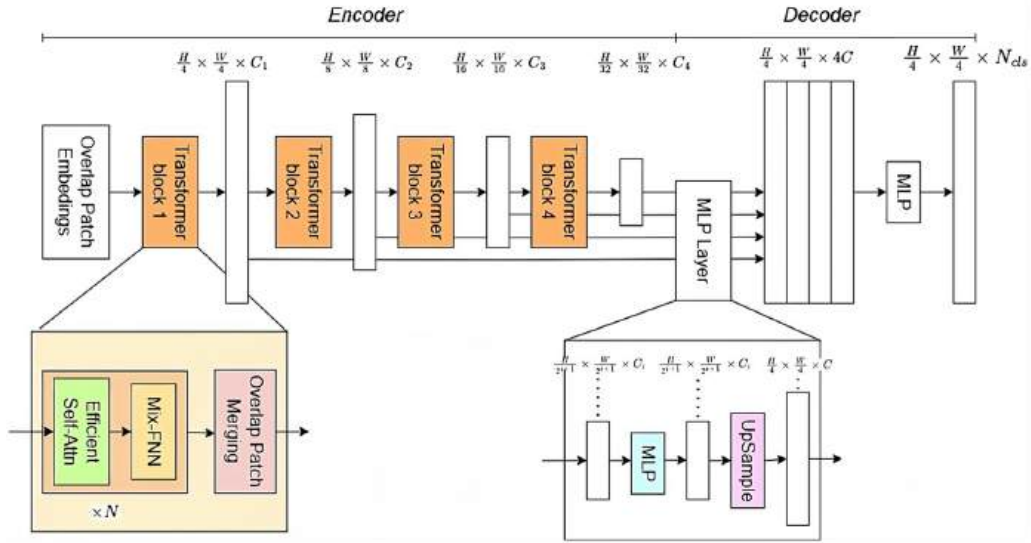


Fig. 8. Segformer architecture.

Precision reflects the model's accuracy in identifying potholes by calculating the proportion of true positives over the total predicted positives. A high Precision value means the model makes few errors in incorrectly classifying areas as potholes, minimizing false alarms. Equation (3) describes how to calculate the precision.

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

The F1 score is the harmonic mean between Recall and Precision, useful when a balance between the two is needed. A high F1 Score indicates the model achieves a good balance between detecting potholes and minimizing false alarms, which is crucial for the architecture's effectiveness. The F1-Score equation is presented in (4).

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

The mean IoU measures the overlap between the predicted and actual segmentation by calculating the intersection divided by the union of the two areas. A high mean IoU value indicates that the model segments potholes with high accuracy, showing a good match between the predicted and actual areas. Equation (5) describes how to calculate the precision.

$$Intercept\ of\ Union\ (IoU) = \frac{Area\ of\ Overlap}{Area\ of\ Union} \quad (5)$$

Fig. 9 shows the graphical representation of the metrics used to evaluate the model's performance in object detection. The image illustrates how precision, recall, and intersection over union (IoU) are calculated by comparing the detection areas, overlap, and the actual object, highlighting the relationships between these key metrics for assessing the model's effectiveness.

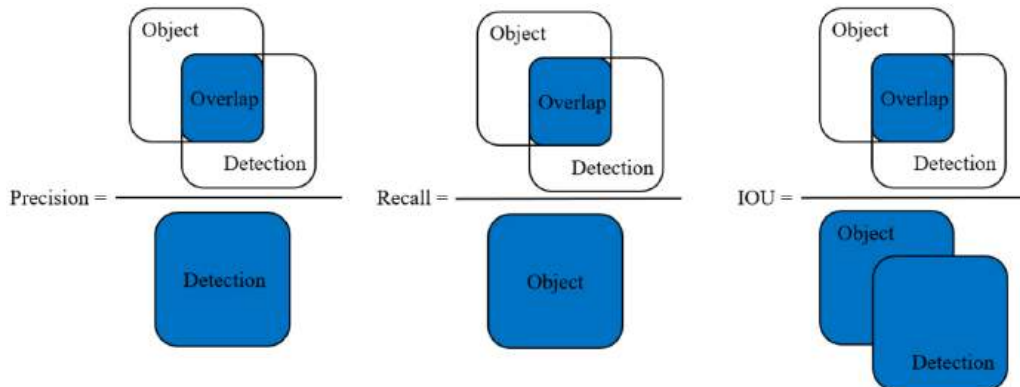


Fig. 9. IOU Evaluation.

Fig. 10 illustrates levels of segmentation quality, represented by the Intersection over Union (IoU) metric. In this figure, the red square corresponds to the segmentation of the pothole predicted by the model, while the blue square represents the manual segmentation or ground truth label. The three examples, Poor, Good, and Excellent, demonstrate the progression of IoU values, where a higher IoU indicates a closer alignment between the predicted and manual segmentations of the pothole.

Validation Loss measures the discrepancy between the model's predictions and the actual values in the validation set. A low Validation Loss value suggests the model generalizes well to new data, which is crucial for its performance in real-world scenarios.

3.4. Data integration

Data integration is a crucial process in our methodology, aimed at merging information obtained from 2D images with 3D point-cloud data. This process allows us to combine the advantages of both data types to achieve more precise pothole segmentation. The process begins by applying preprocessing to the point cloud for noise reduction using the RANSAC (Random Sample Consensus) algorithm. Subsequently, the contour obtained in the pothole segmentation stage is used to delineate the same area in the corresponding point cloud.

3.4.1. Point cloud data preprocessing

After collecting the point-cloud data, preprocessing is needed when you consider that while scanning a scene in actual life, the captured clouds include snippets of the scene. Therefore, alignment to generate a complete map of the scanned environment is mandatory.

Given the non-uniform nature of the road surfaces, aligning them with a common coordinate system is a crucial step. This alignment ensures uniformity across all samples, allowing for reliable and accurate analyses. The RANSAC algorithm (Fischler et al. [37]), a reliable tool, is used to identify a plane that best fits the terrain surface in the point cloud.

The process starts with plane segmentation, where RANSAC randomly selects subsets of points and fits a plane model, calculating the parameters (normal and distance to the origin) of the plane that best fits the selected points. Equation (6) represents the plane, where A , B , C are the components of the normal vector to the plane, and D is the distance from the origin to the plane along the normal vector. The algorithm then evaluates how many points are within a specified distance threshold, selecting the model that maximizes the number of inliers as the final plane model. Finally, the rotation vector needed to align the normal of the segmented plane with the Z-axis is calculated and applied to the point cloud, leveling the plane.

$$Ax + By + Cz + D = 0 \quad (6)$$

Here (A, B, C) is the normal vector to the plane and D is the distance from the plane to the origin.

Once the point cloud is aligned, the next step in preprocessing is the removal of outliers that may have been introduced by reflective surfaces or high illumination during the capture process. Removing these points is essential to retain only the relevant data in the point cloud. This cleaning process is performed using the Nearest-Neighbor Filtering method. This method compares a point's distance to its nearest neighbors. If a point's distance to its neighbors is significantly greater than the average distance of other points to their neighbors, that point is considered an outlier and is removed. This filtering ensures that only the points representative of the terrain and the potholes are retained for the depth estimation.

3.4.2. Point cloud segmentation

The preprocessed point clouds are integrated with the pothole detections obtained from the images. Each image is linked to a point cloud, providing a three-dimensional representation of the surface where the pothole was detected. After storing the coordinates during SegFormer detection, this architecture converts these two-dimensional image coordinates into three-dimensional spatial coordinates. This conversion process uses the intrinsic camera parameters as the basis for calculating the scale factor, allowing the precise localization of the pothole in 3D space.

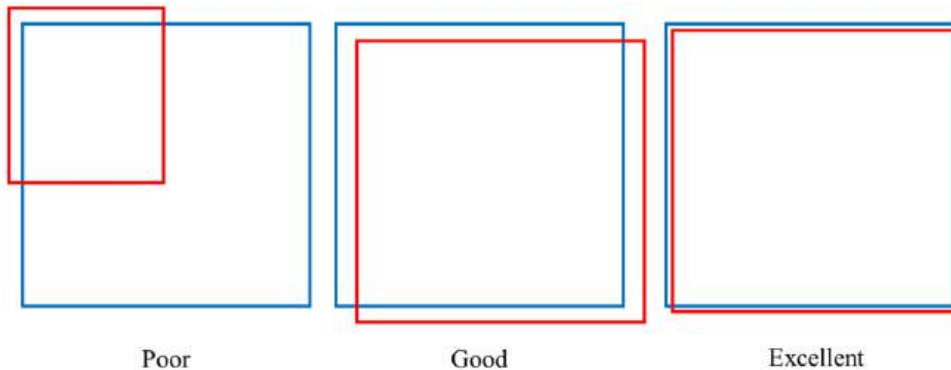


Fig. 10. IOU Evaluation.

The scale factor in computer vision is used to convert 2D coordinates from an image to 3D coordinates. This process is essential when working with images projected onto a plane, as in the case of cameras, and when it is necessary to estimate the depth or spatial location of detected objects. The process of calculating the scale factor Z is described below.

The camera is usually modeled as a pinhole camera, where the relationship between the 3D coordinates of a point (X, Y, Z) and its 2D projections in the image (u, v) is given by the following matrix Equation (7):

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{z} \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad (7)$$

Here, f_x and f_y are the focal lengths in pixels, and c_x and c_y are the offsets of the optical center (or the coordinates of the image center).

To obtain the scale factor, the distance from the object to the camera is used in this case, the distance from the camera to the pavement, and it is calculated using Equation (8).

$$Z = \frac{f_x \cdot (X - c_x)}{u} = \frac{f_y \cdot (Y - c_y)}{v} \quad (8)$$

Once the scale factor Z is obtained, it is used to recover the coordinates X and Y in three-dimensional space by multiplying the image coordinates u and v by Z .

After the exact location of the pothole in the point cloud is established, the architecture continues to retain only the points that belong to the damaged areas, as shown in Fig. 11. This step allows working only with the points of interest, preventing cases where more than one deterioration is found in the same capture and reducing the storage space required for the point clouds.

3.4.3. Parameters estimation

According to the FHWA manual [35], the diameter of a pothole is defined as the diameter of the largest circle contained within the contour of the pothole. We apply an iterative approach to identify the largest possible circle that can be inscribed within the detected contour of the pothole to calculate the diameter of a pothole. This process begins by analyzing the images where the pothole has been detected, progressively drawing circles of different sizes until the largest circle that fits completely within the edges of the pothole's contour is found. The diameter of this circle is considered to be the diameter of the pothole.

Subsequently, it is necessary to find a scale factor to convert distances from pixels to meters. This scale factor is calculated using the camera's intrinsic parameters, such as the field of view (FOV), image resolution, and capture height.

First, the real width (W_{real}) and real height (H_{real}) of the camera's field of view are calculated. These dimensions are determined using the capture height ($H_{capture}$) and the horizontal ($\theta_{horizontal}$) and vertical ($\theta_{vertical}$) angles of view. The following equations (9) and (10) are derived from the geometry of the triangle formed between the camera and the edges of the field of view.

For the real width:

$$W_{real} = 2 \times H_{capture} \times \tan\left(\frac{\theta_{horizontal}}{2}\right) \quad (9)$$

For the real height:

$$H_{real} = 2 \times H_{capture} \times \tan\left(\frac{\theta_{vertical}}{2}\right) \quad (10)$$

The next step is to calculate the horizontal and vertical scales. These are obtained by dividing the real width and height of the field of view by the horizontal (Res_{width}) and vertical (Res_{height}) resolutions in pixels, respectively.

The horizontal scale (H_{scale}) is calculated as:

$$H_{scale} = \frac{W_{real}}{Res_{ancho}} \quad (11)$$

The vertical scale (V_{scale}) is calculated as:



Fig. 11. Segmentation.

$$V_{scale} = \frac{H_{real}}{Res_{alto}} \quad (12)$$

To convert the radius of a circle measured in pixels (*Diameter_pixels*) to meters (*Diameter_metros*), the horizontal scale is used. This is shown in Equation (13).

$$Diameter_{metros} = Radio_{pixels} \times H_{scale} \quad (13)$$

We used the segmented pothole and its corresponding leveled point cloud for the depth estimation, having previously applied a filtering process to remove outliers. As illustrated in Fig. 12, the procedure begins with estimating the distance from the sensor to the terrain's surface. This step is carried out by calculating the mode of the Z distances of the points in the point cloud, thus representing the distance to the pavement surface. Then, the farthest point from this surface in the point cloud is identified. The pothole's depth is determined by obtaining the difference between the distance from the sensor to the pavement surface and the distance to the furthest point from the surface level, as shown in Equation (14).

$$Depth = Z_{Surface} - Z_{(Deepestpoint)} \quad (14)$$

3.5. Fuzzy model evaluation for road potholes

Individually detected potholes are evaluated using a fuzzy logic model, which allows handling of the uncertainty present when determining the severity of the damage, for instance, the measurement process and the quantification of the damage. Traditional road damage assessment is subject to inspector criteria, which can lead to inconsistencies.

After consulting deterioration evaluation manuals, such as those from the FHWA [35] and SICT [26], it was determined that the key parameters for characterizing a pothole and determining its severity are depth and diameter. Therefore, these same variables are used as inputs for the fuzzy model. As for the output variables, there are two: the first corresponds to severity, which is also interpreted as the urgency of repair, and the second is the recommendation for the repair technique. We choose as the inference engine the Mamdani model, described in Equation (15), for its high interpretability and ability to handle uncertainty.

$$z^* = \frac{\sum vCT(z)z}{\sum vCT(z)} \quad (15)$$

Figs. 13 and 14 show the input and output membership functions, respectively. We use trapezoidal functions to provide a transition between each of the fuzzy sets, as these functions provide a more realistic representation of the bump decay process. The choice of trapezoidal functions over other membership functions (such as triangular or Gaussian) is justified for two main reasons: first, they allow modeling ranges of values with a full degree of membership (upper plateau of the trapezoid), which better reflects the intervals where deterioration remains stable in a specific category; second, they provide gradual transition zones (slopes of the trapezoid) that more naturally represent the progression of deterioration between different levels of severity. This feature is significant in evaluating potholes, where the change between deterioration states does not occur abruptly, but rather there is a gradual process of degradation.

Data for each input variable were normalized using the min-max equation (16) to ensure that both variables are on the same scale. The limits for each input variable are described in Table 2. Normalization of input data allows the fuzzy model to process the data regardless of the original scale, helping to standardize the data and reduce the impact of differences in the scale and magnitude of data attributes, avoiding biases caused by differences in magnitude.

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (16)$$

We generate fuzzy rules to determine the urgency/severity of the attention needed to repair the pothole and the maintenance

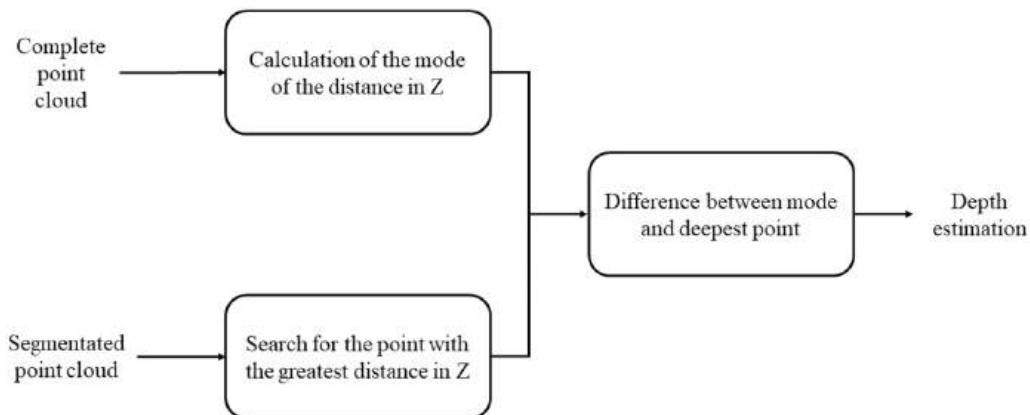


Fig. 12. Depth estimation.

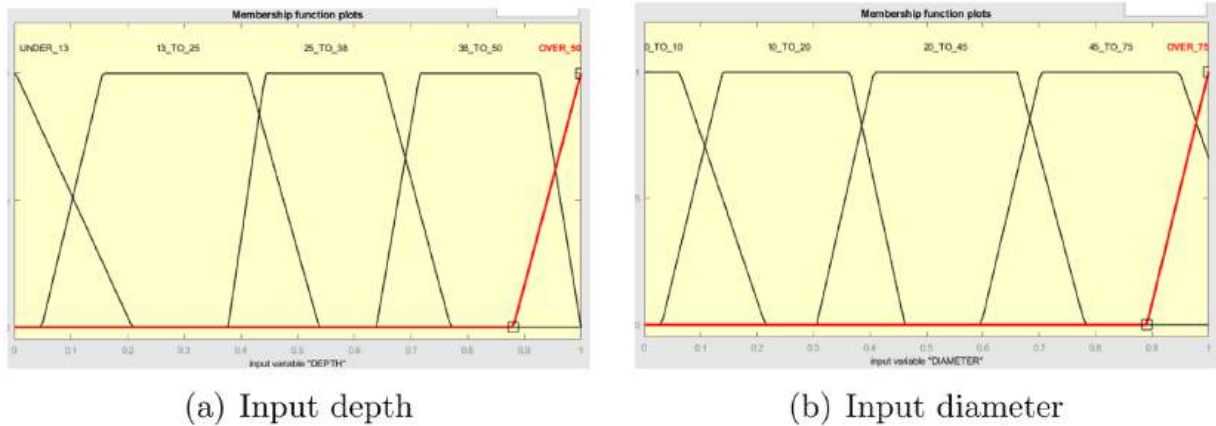


Fig. 13. Input membership functions.

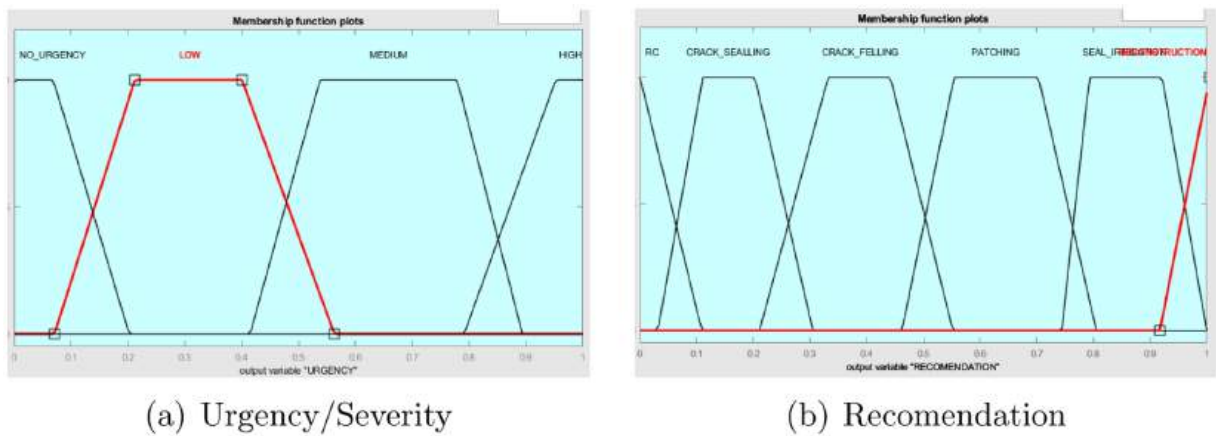


Fig. 14. Output membership functions.

recommendation. The Mandami model is applied considering Equation (15). In Table 3, we describe the fuzzy rules of the system, which were carefully designed with a preventive maintenance approach to determine the most cost-effective time of intervention and the extension of the remaining pavement life. The rules take into consideration the inspection manuals proposed by FHWA [35] and SICT [26] that identify depth as the key parameter influencing severity and consider all combinations of diameter and depth to address the full range of pothole evaluation.

The surface shown in Fig. 15 illustrates the relationship between the diameter and depth of the potholes and the severity/urgency of their repair. Likewise, in manuals such as [35] and [26], we have considered in the rules description that depth significantly impacts determining repair urgency, shown in Fig. 15 a). This is because a greater depth usually indicates more severe structural damage to the road, necessitating a faster intervention. The graph helps to visualize how various combinations of diameter and depth influence the urgency of repair, emphasizing that depth is the primary factor in assessing the severity of damage. Fig. 15 b) illustrates how repair recommendations do not depend solely on severity, but consider the interaction between the diameter and depth of the pothole.

Table 2

Inputs and outputs variables of the fuzzy model.

INPUTS		OUTPUTS	
DIAMETER (cm)	DEPTH (mm)	SEVERITY / URGENCY	RECOMENDATION
MINIMAL (Under 10)	MINIMAL (under 13)	NO DAMAGE	ROUTINE MAINNENTENCE
LOW (10–20)	LOW (13–25)	LOW	CRACK SEALLING
MEDIUM (20–45)	MEDIUM (25–37)	MEDIUM	CRACK FEALLING
MODERATE (45–75)	MODERATE (37–50)	HIGH	PATCHING
HIGH (Over 75)	HIGH (over 50)		SEAL IRRIGATION
			RECONSTRUCTION

Table 3

Fuzzy rules of the pothole evaluation model.

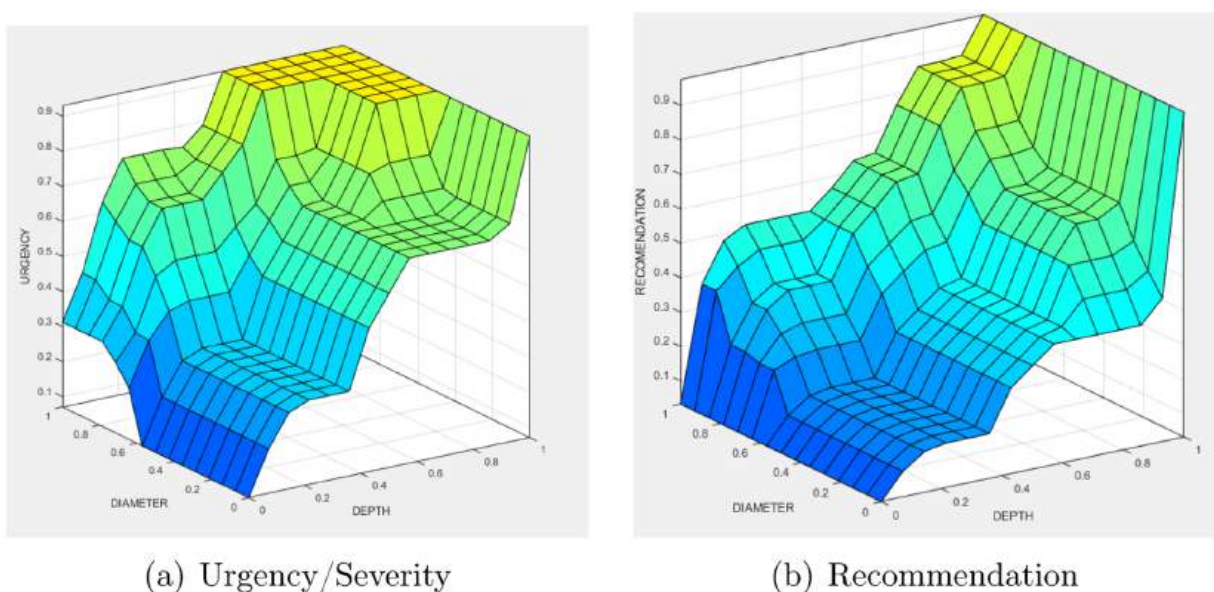
INPUTS			OUTPUTS	
DEPTH		DIAMETER	URGENCY/SEVERITY	RECOMMENDATION
MINIMAL	and	MINIMAL	NO DAMAGE	ROUTINE MAINTENANCE
MINIMAL	and	LOW	NO DAMAGE	ROUTINE MAINTENANCE
MINIMAL	and	MEDIUM	NO DAMAGE	ROUTINE MAINTENANCE
MINIMAL	and	MODERATE	LOW	ROUTINE MAINTENANCE
MINIMAL	and	HIGH	LOW	ROUTINE MAINTENANCE
LOW	and	MINIMAL	LOW	CRACK SEALLING
LOW	and	LOW	LOW	CRACK SEALLING
LOW	and	MEDIUM	LOW	CRACK FELLING
LOW	and	MODERATE	MEDIUM	CRACK FELLING
LOW	and	HIGH	MEDIUM	PATCHING
MEDIUM	and	MINIMAL	MEDIUM	CRACK SEALLING
MEDIUM	and	LOW	MEDIUM	PATCHING
MEDIUM	and	MEDIUM	MEDIUM	PATCHING
MEDIUM	and	MODERATE	HIGH	PATCHING
MEDIUM	and	HIGH	HIGH	PATCHING
MODERATE	and	MINIMAL	MEDIUM	CRACK SEALLING
MODERATE	and	LOW	MEDIUM	PATCHING
MODERATE	and	MEDIUM	HIGH	PATCHING
MODERATE	and	MODERATE	HIGH	SEAL IRRIGATION
MODERATE	and	HIGH	HIGH	SEAL IRRIGATION
HIGH	and	NONE	HIGH	RECONSTRUCTION

Elevated regions (in green-yellow) indicate where more thorough interventions are recommended, while blue areas suggest less intensive treatments.

4. Results

This section presents the results obtained through the application of the proposed architecture. Data derived from sample collection, the training of the semantic segmentation model, depth estimation, and damage assessment are shown. In addition, visual examples of the detections and segmentations performed are included, accompanied by tables and graphs that illustrate the evaluation metrics used, such as the F1 score, recall, and precision, as well as the results of the fuzzy logic system applied to the evaluation of 11 potholes.

Fig. 16 shows how manual depth measurements of potholes are taken. A rigid flat metal is used as a reference to establish the surface plane positioned on two support points along the pothole's edge. Then, a digital caliper is used, with its depth rod positioned at various points within the pothole until the deepest point is found.

**Fig. 15.** Surfaces.

4.1. Dataset

The results of the collection, cleaning, and categorization of images led to a dataset consisting of a total of 583 images selected to represent various conditions under which potholes can be found. When selecting the images, the goal is also to balance the number of images that contain potholes with those that do not, ensuring that both groups have a similar quantity. Table 4 shows both the number of images obtained and the number of selected images that contain environmental or external noise. Particularly, the “other” column refers to images that contain a combination of stains and shadows, paint on the pavement, manhole covers, high contrast caused by lighting, or elements such as trash or foreign objects on the pavement. This category also includes patches from previous repairs or variations in pavement color. We selected these images from different contexts to provide a comprehensive foundation for training models to detect potholes accurately, even under adverse or complex environmental conditions.

We use the imbalance ratio (IR) to measure the balance of the classes. Equation (17) is applied, where N_- is the number of images in the minority class and N_+ is the number of images in the majority class.

$$IR = \frac{N_-}{N_+} = \frac{234}{299} = 0.78 \quad (17)$$

Determining an IR value to know when a dataset is unbalanced is not well established. However, Lango [38] has proposed that an IR greater than 1.5 represents an unbalanced data set. On the other hand, Tsai et al. [39] explained that if the IR value is higher than ten, it is difficult to generate an effective classifier. These can also be extrapolated to the context of supervised segmentation. Hence, the IR value obtained represents, in our case, that the generated pothole dataset is balanced.

Fig. 17 shows images from the collected dataset. The samples in incise a) and b) display images of the road with some shadows. Samples c), e), and f) show images with potholes, delineating their contours. In e) and f), the shadows partially or fully cover the potholes. These examples show the different types of images provided to the model to correctly identify and detect potholes, even in the presence of shadows or stains.

Fig. 18 shows an example of a point cloud in which a pothole was captured. Fig. 18a shows the RGB point cloud on the pavement surface. Fig. 18b shows the colored point cloud that shows the areas of the surface farthest from the sensor. Fig. 18c shows the figure of the pothole from a planar view; here, we can see the depth of the pothole.

4.2. Segformer training

Fig. 19 presents three graphs showing the metrics of the validation set for the trained model. Fig. 19a shows the evolution of the Mean IoU (Intersection over Union), which increases steadily, indicating an improvement in the model's segmentation accuracy. Fig. 19b shows the evolution of the mean accuracy, which also increases but with fluctuations, suggesting an improvement in the classification of pixels. Fig. 19c shows the progressive decrease in validation loss (val_loss), indicating that the model is learning and improving its generalization ability. Fig. 20.

Fig. 21 shows the confusion matrix obtained during the validation phase of the SegFormer model. The matrix indicates that the model achieves 99 % accuracy in correctly classifying the background, with only a 1 % error in confusing the background with potholes. On the other hand, the model identifies potholes with an accuracy of 83 %, although it makes an error of 17 % by incorrectly classifying them as background.

Fig. 21 shows the confusion matrix for the test set. During this phase, the model demonstrates an accuracy of 100 % to correctly identify the background without making any errors in its classification. Regarding the potholes, the model has an accuracy rate of 80 %, mistakenly classifying 20 % of the potholes as background.

Table 5 summarizes the evaluation metrics following the model's training. The key results include a recall of 90.87 %, a precision



Fig. 16. Depth measurement sample.

Table 4
Dataset Results.

CLASS	IDEAL CONDITIONS	SHADOWS	IMAGES STAIN	OTHER	SIZE (pixels)	TOTAL IMAGES
POTHOLES	70	67	26	136	512 X 512	299
NOT POTHOLES	43	37	30	124	512 X 512	234

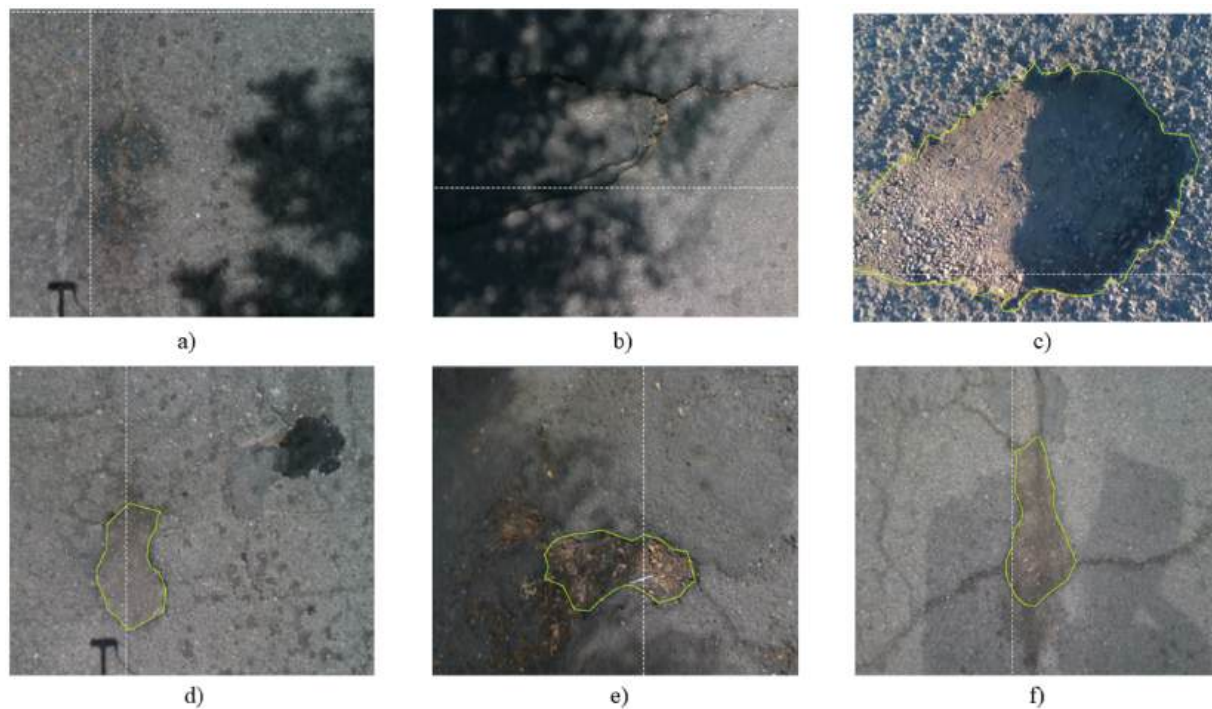


Fig. 17. Examples of image annotated in the dataset.

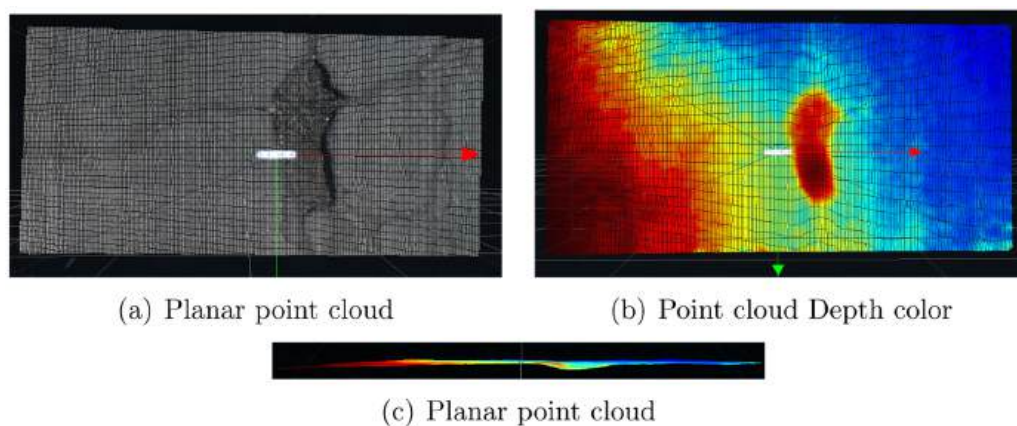


Fig. 18. Point cloud example.

of 90.01 %, an accuracy of 86.8 % a mean IoU of 85.872 %, and a loss of 0.0431. These results demonstrate that the model effectively detects actual potholes, as reflected in a high recall, indicating its ability to identify potholes correctly. Simultaneously, the precision achieved indicates the low rate of false positives. The F1 score of 90.433 % confirms the good balance and overall effectiveness of the model, making it suitable for practical applications. Additionally, an IoU of 85.872 % highlights the model's ability to accurately segment the areas of interest in the images. Finally, the low loss (0.0431) indicates that the model has learned well and is generalizable

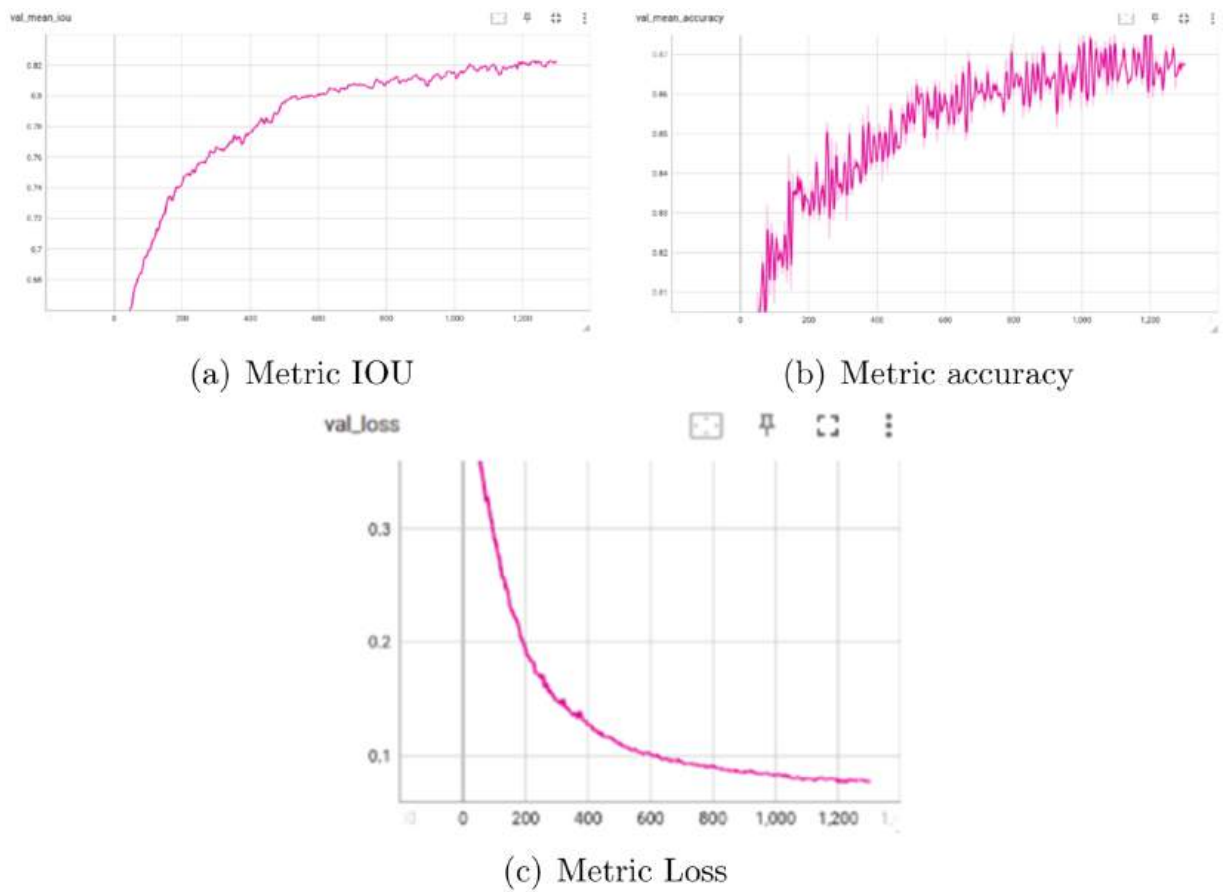


Fig. 19. Training metrics.

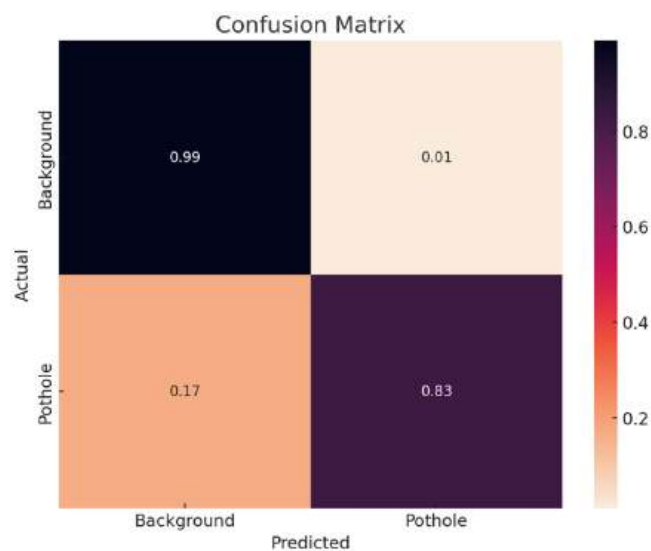


Fig. 20. confusion matrix Validation Dataset.

effectively, which is essential for its performance on unseen data.

Fig. 22 illustrates the results of semantic segmentation under various conditions, organized into four sets of images arranged in rows a), b), c), and d). In each set, column A) shows the original image input to the model, column B) presents the ground-truth mask,

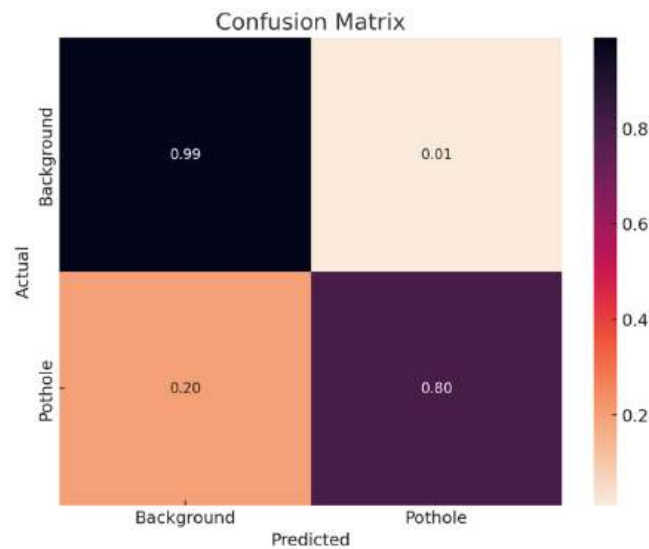


Fig. 21. Confusion matrix Test Dataset.

Table 5
Metrics results.

Metric	Value
Recall	90.87 %
Precision	90.01 %
Accuracy	86.8 %
F1 Score	90.433 %
IOU	85.872 %
Loss	0.0431

corresponding to the manual labeling of the image, and column C) displays the prediction generated by the inference model. In row a), the model correctly identified a pothole partially covered by a shadow. In row b), a pothole is also shown, but under normal conditions, it is similarly recognized with accuracy. In row c), a manhole without the presence of a pothole is shown, and the model appropriately does not identify it as such. Finally, in row d), a cracked pavement is shown alongside a part of a manhole, and the model again makes a correct prediction by not identifying these features as a pothole.

The model currently detects potholes accurately, even in the presence of shadows, stains, manholes, or multiple potholes in the same image (Fig. 23). However, there is room for improvement in more complex situations like pothole overlapping or the presence of soil/water filling. These limitations arise from the model's difficulty in establishing pothole boundaries. Prior work like [13] also has indicated this as an area of opportunity.

4.3. Case of study

The case study selected to evaluate the architecture proposed in Fig. 1 is a street located in the city of Culiacán, Sinaloa, Mexico. Fig. 24 shows the streets highlighted in blue, where the architecture tests were carried out. The length of the street is 800 m, where 11 potholes were identified.

4.4. Results of the case of study

Table 6 provides a detailed view of the results obtained from analyzing 11 pothole samples. The results indicate that the architecture achieves a high level of accuracy in estimating both the diameter and depth of potholes. The statistical results of the comparison are also shown, including the minimum, maximum, and average errors. For diameter estimations, the minimum error is 0.7 cm, the maximum error is 10.7 cm, and the average error is 2.14 cm. For depth estimations, the minimum error is 1.2 mm, the maximum error is 16.22 mm, and the average error is 5.94 mm. These metrics demonstrate that the architecture provides reliable estimates that are close to the real measurements.

In addition, the severity evaluation and repair suggestions generated by the fuzzy logic system reveal that the section analyzed exhibits moderate to high deterioration levels. The suggestion of patching as the optimal repair method underscores the practicality of adopting this approach for preventive and corrective road maintenance. Overall, the research emphasizes the effectiveness of the

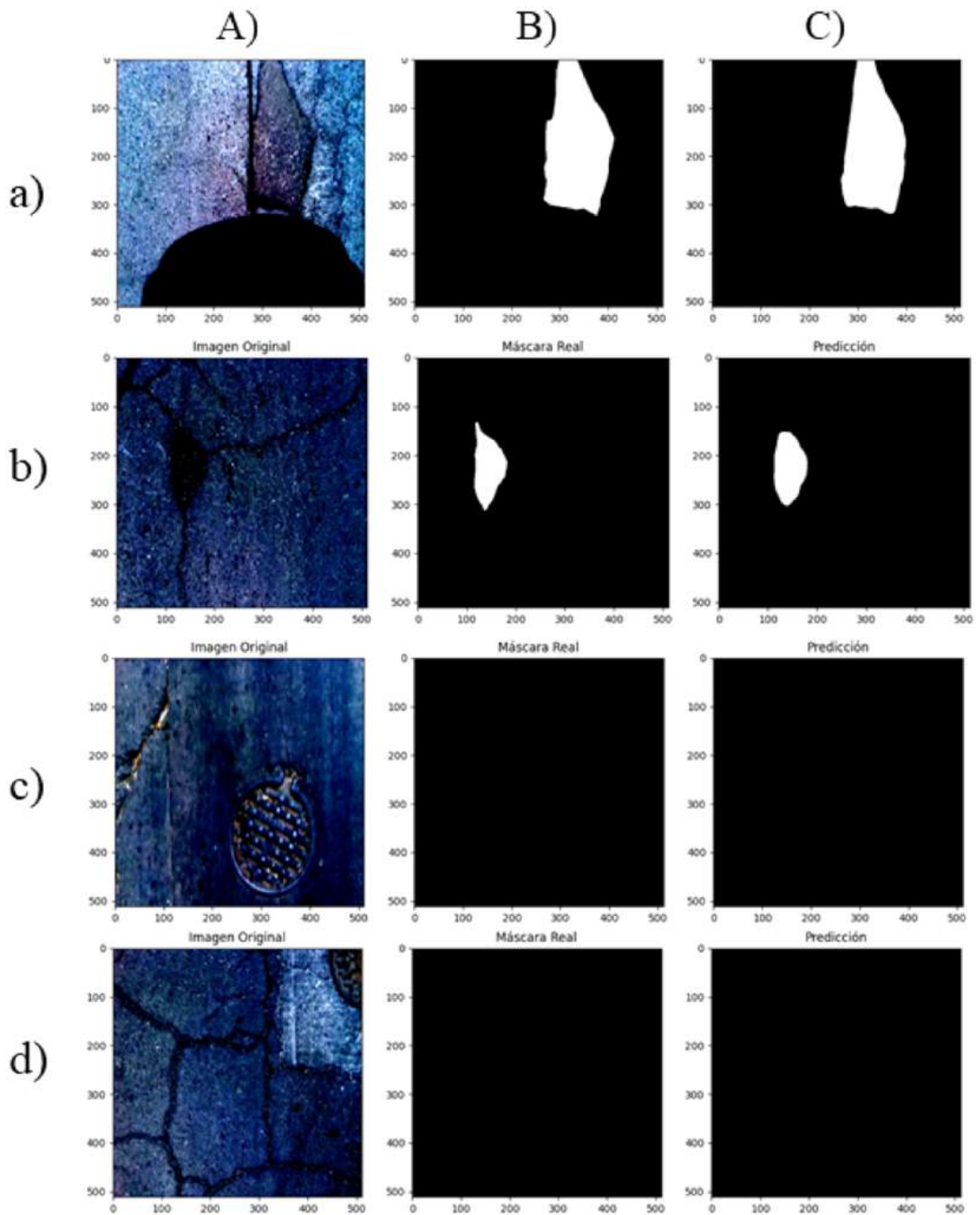


Fig. 22. Sematic Segmentation results.

suggested architecture as an accurate and efficient instrument for road maintenance management, providing a strong alternative to conventional manual approaches. Fig. 25.

The architecture exhibits a minimum error of 1.2 mm, a maximum error of 16.22 mm, and an average error of 5.94 mm. This

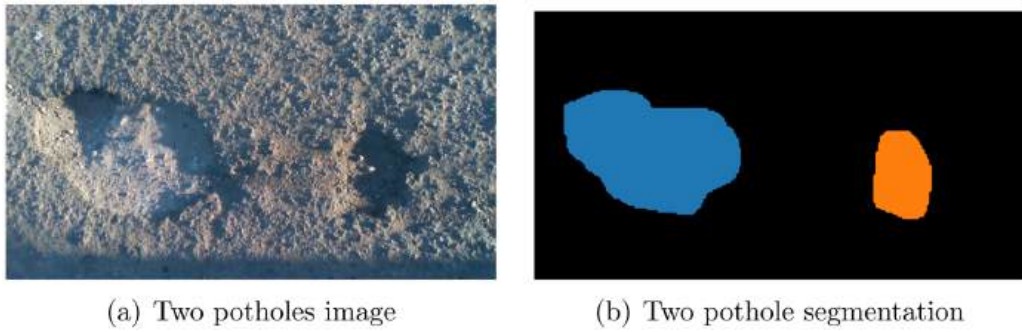


Fig. 23. Segmentation of two potholes in the same image.

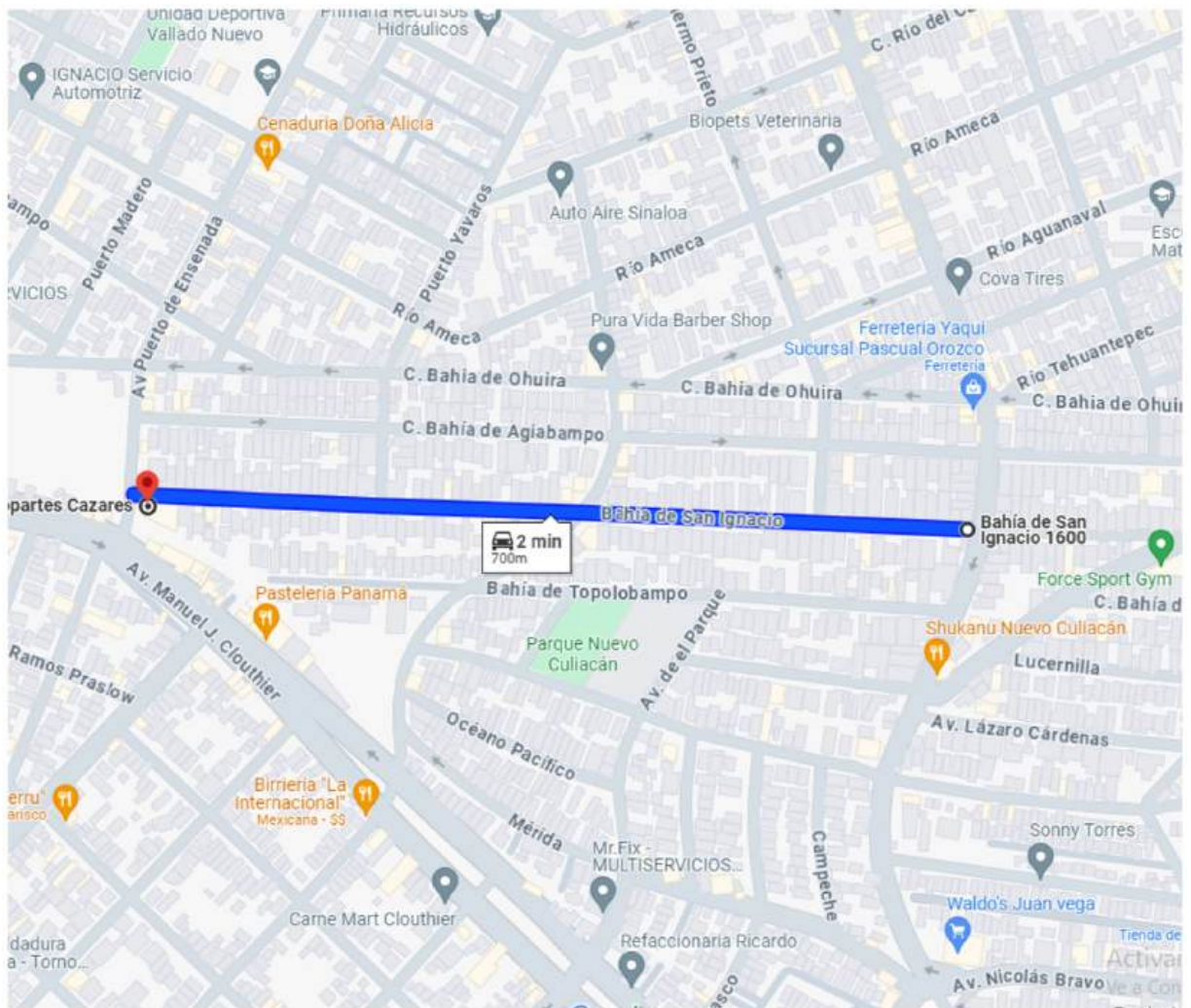


Fig. 24. Localization of the case study.

indicates that the depth estimation error is low in most situations, less than 1 centimeter.

5. Discussion

Table 7 and Table 8 compare various state-of-the-art works, highlighting costs, detection errors, limitations, and depth errors for

Table 6
Case of study results.

Pothole	Diameter (cm)		Depth (mm)		Fuzzy Logic	
	Real	Estimate	Real	Estimate	Severity	Recommendation
1	38	38.88	26	20.33	Medium	Patching
2	64	63.08	25	30.58	Medium	Patching
3	25	18.51	37.5	47.36	Medium	Patching
4	40	43.80	41.5	34.95	High	Patching
5	32	33.21	48	43.54	Medium	Patching
6	30	28.37	26	24.8	Medium	Patching
7	80	69.30	30	35.08	High	Patching
8	86	80.75	26	31.84	Medium	Patching
9	39	38.30	45	28.78	High	Patching
10	70	64.26	46	37.39	High	Patching
11	72	68.25	26	34.16	Medium	Patching
Diameter (cm)			Depth (mm)			
Error Min		0.7	Error Min		1.2	
Error Max		10.7	Error max		16.22	
Average		2.14	Average		5.94	

pothole detection. Previous works like Bosurgi, G. et al. [17], and Wu, H. et al. [18] are significantly more expensive solutions, priced at \$1,000,000 and \$10,000 respectively, compared to the architecture presented in this work, which costs only \$500. Although HiProSS offers an extremely low depth error of 1 mm and the SICK LMS 511 has an error range of 15–28 mm, both works have considerable limitations, such as the high cost of HiProSS and the dependence of the SICK LMS 511 on weather conditions and pavement geometry. On the other hand, while having an average error of ± 10 mm and operating within a speed range of 10–50 km/h, our architecture proves to be a more cost-effective and versatile alternative.

Another aspect worth highlighting is the detection results. Although there are works with higher percentages in metrics such as accuracy, recall, and F1 score, like Fan, R. et al. [9], or Bosurgi, G. et al. [17], the performance of our architecture is competitive and functional in most cases. In addition, we provide the severity and urgency to carry out the repair and maintenance recommendation for the potholes, which overcomes all the work presented in Table 8.

When comparing depth errors, the HiProSS system [17] offers the highest precision with an error of 1 mm, but its high cost of \$1,000,000 makes it inaccessible for many projects. The system by Wu, H. et al., [18] is more affordable at \$10,000 but presents a higher error. In comparison, Ahmed's work, though low-cost, shows considerable variability in depth calculations, especially at higher speeds, where it can reach up to 26.6 % at 20 km/h. In contrast, our work maintains a more consistent error of ± 7.82 mm and operates efficiently within a speed range of 10–50 km/h, offering a more balanced and reliable option for various applications.

Our work stands out for its balanced approach that combines cost-effectiveness, detection accuracy, and a unique capability for depth estimation, severity/emergency assessment, and repair recommendations. This positions our work as a singular option compared to manual work and also compared to the studies shown in Table 8. Additionally, our work benefits the institutions



Fig. 25. Visual representation of pothole diameter.

Table 7
Technical comparison.

Article	Detection technique	Evaluation	Device	Cost (USD)	Depth estimation	Mode
G. et al. [17]	Digital image processing	Severity	HiProSS	\$1,000,000	Yes	Dynamic
H. et al. [18]	CNN	No apply	Laser SICK LMS 511 Panoramic camera	\$10,000	Yes	Dynamic
M. R. et al. [30]	Point cloud segmentation	Severity	2 Laser Riegl VW-450	Not specified	Yes	Dynamic
R. et al. [9]	Disparity maps	No apply	Zed Camera	\$449	Yes	Not specified
Peralta-Lopez, et al. [28]	CNN	No apply	Zed Camera	\$449	No	Dynamic
A. et al. [27]	Fotogrametry	No apply	Stereo Camera sensor laser	Low-cost	Yes	Dynamic
I. et al. [29]	Transformer	No apply	Zed Camera	\$449	No	Dynamic
R. et al. [13]	3DpredictNet	No apply	3D GO! SCAN	\$20,000	Yes	Static
This work	Transformer	Severity \Urgency Tecnique repair	RealSense D435if	\$500	Yes	Dynamic

Table 8
Result comparison.

Article	Detection results	Area under analysis	Depth and Area Error	Velocity	Limitations
G. et al. [17]	Precision 89.75 % Recall 92.95 % F1 Score 91.28 %	1 m x 1.2 m	Depth 1 mm Area min 5.15 % max 8.88 %	Traffic speed	Cost Only performs depth validation in the laboratory
H. et al. [18]	Mean accuracy detection 1.5–2.8 cm	No mention	Length and Width 15–28 mm	40 km/h	Weather Blocking of potholes Geometry of pavements Only well-defined round potholes Does not compare actual depth with estimates in potholes.
M. R. et al. [30]	Not specified		max 46 mm	10 km/h	Velocity low Cost
R. et al. [9]	Accuracy 98.7 % Pixel level accuracy 99.6 %	No mention	Do not apply	Not specified	Only 2D Data No measurement estimation
Peralta-Lopez, et al. [28]	F-Score 82.6 % M IOU 70.3 % Runtime 34.7 ms	No mention	Do not apply	Not specified	Only 2D Data No measurement estimation
A. et al. [27]	Not specified	No mention	Depth error Static 5.3 % 10 km/h 7.9 % 15 km/h 14.4 % 20 km/h 26.6 %	10 km/h 15 km/h 20 km/h	Only 5 potholes in the tests 26 % error on 20 km/h
I. et al. [29]	Accuracy 96.21 % Precision 83.07 % Recall 83.07 % F1 Score 77.24 % IOU 62.91 %	No mention	Do not apply	Not specified	Only 2D Data No measurement estimation
R., et al. [13]	Accuracy 97.63 % Precision 94.83 % Recall 92.61 % F1 Score 93.07 % IOU 88.38 %	No mention	Volume error 13.94 %	Static	Pothole complex shapes Surface irregularities Pothole obscured
This work	Recall 90.87 % Precision 90.01 % F1 Score 90.433 % IOU 85.872 % Loss 0.0431	2.47 m x 2.44 m Area 3.56 m ²	Depth +−5.94 mm Diameter 2.14 cm	10–50 km/h	Changes in illumination Pothole obscured Speed over 50 km/h

responsible for road maintenance management in urban areas by reducing inspection times and costs through early-stage damage repairs.

Table 8 displays a comparison of the detection results, the errors in estimating diameter, area, and depth, as well as the limitations of multiple studies, including the newly proposed method. An important metric is the Intersection Over Union (IOU), which evaluates the overlap between predicted and actual pothole regions. A higher IOU suggests improved accuracy in detecting the pothole's contour, leading to more precise measurements of its dimensions. The Intersection over Union (IOU) plays a critical role in determining the diameter and area of a pothole. Although certain studies only focus on calculating the area without explicitly estimating the diameter or depth, a low IOU adversely affects the accuracy of the pothole's contour detection. This consequently increases the error in the estimation of both the area and diameter. On the other hand, our proposed method achieves an IOU of 85.872 %, facilitating reliable diameter measurements with an average error of ± 2.14 cm. Unlike studies with lower IOU values, such as Katsamenis et al. (62.91 %), our achieved IOU rate improves diameter estimation, where incomplete pothole detection could lead to underestimated area and diameter measurements. An important point to emphasize is the area of data capture; if this is less than the width of a lane, it is necessary to travel the lane more than one time in order to capture its total width and observe its condition in its entirety. As shown in Table 8, most of the work does not mention these data.

Fig. 26 shows a visual comparison of our work against previous studies of the metrics related to the detection models of each paper. In comparison with other works, our detection model has room for improvement; however, it should be noted that our work not only performs pothole detection but also an evaluation based on the estimation of its diameter and depth to provide a repair recommendation. It should also be noted that higher IOU work such as [13] does not have in-motion testing, and its main focus is detection and volume acquisition.

Our experimental analysis identified operational constraints in high-speed and large-scale deployment scenarios. First, at velocities exceeding 40 km/h, the depth sensor exhibits degraded reliability due to frame rate limitations, which blur the image capture, causing distortion and making it impossible to recognize it. Second, storage demands inherently limit scalability. The system generates approximately 5 GB/min of multimodal data (point clouds and images), requiring high storage capacity for prolonged inspections. Furthermore, while our architecture mitigates common occlusion issues by positioning the camera orthogonally to the pavement surface—avoiding angular obstructions—it remains susceptible to environmental factors. Like most vision-based systems, performance degrades under adverse weather (e.g., rain, snow) or in conditions such as glare, as these phenomena alter surface reflectivity and distort sensor measurements. These findings underscore the need for adaptive sensor fusion strategies and edge-compatible compression algorithms to enhance robustness in real-world deployments.

Maintenance planning can be enhanced through this architecture by obtaining comprehensive road condition data and identifying homogeneous zones requiring repair. This enables informed decision-making regarding appropriate repair techniques for each zone based on objective severity assessments.

The early detection of potholes allows for the application of less costly repair techniques, preventing further deterioration and avoiding more expensive interventions in the future. By determining repair activities through collected data, resources are optimized, and the remaining service life of roads is extended through timely and appropriate interventions.

6. Conclusions

This paper aims to contribute to generating an automated inspection that quantifies and evaluates potholes in urban streets by integrating image data and point clouds. The contributions of this study effectively address the limitations identified in the state of the art, particularly in false positive detection and the lack of repair recommendations based on damage severity. Besides, we provide a fuzzy decision support system that recommends the maintenance technique that repairs the damage for each pothole detected.

The sensor used demonstrated excellent performance in most scenarios, achieving an average error of 6 millimeters in depth estimation. However, in cases where potholes contain water, the total depth is not precisely detected. Additionally, under abrupt changes in lighting conditions, the sensor takes approximately one second to focus properly.



Fig. 26. Training result comparison.

We build a dataset that includes variations in environmental conditions, such as shadows, manhole covers, and stains, to give the model real-world scenarios. In addition, we implemented transfer learning in the Segformer network to segment potholes, demonstrating adequate performance and achieving a precision of 90.01 % and a mean IoU of 85.87 %. These results highlight the architecture's ability to operate efficiently under varied conditions, surpassing the limitations of other works such as [34] and [9], which rely on more controlled environmental settings. Additional improvements for the SegFormer model could be considered, such as hybridization with attentional mechanisms or reinforcement learning, but they were not part of the study's main objective, which is the evaluation and recommendation of repair techniques from an integration of 2D and 3D data.

The Segformer model, through transfer learning, achieves a high Intersection over Union (IOU), which guarantees accurate pothole detection and allows for precise measurement of their diameters with minimal error. This result underscores the importance of a high IOU for reliable pothole dimensioning, as lower IOU values in other studies often lead to underestimated areas and diameters due to incomplete detection. However, the model has areas for improvement, particularly in scenarios where potholes are not fully identified. This occurs in cases where potholes have been partially filled, resulting in the edges being obstructed.

Identifying potholes in the images is a fundamental aspect that enables data integration in this work. This integration is key to effectively segmenting the area corresponding to the pothole in both the image and the point cloud, significantly reducing the amount of information that needs to be processed when calculating measurements such as the diameter and depth of the pothole. These measurements, in turn, are the essential input for the fuzzy logic system developed in this study, which not only assesses the severity of the damage but also prioritizes the urgency of the repair and suggests the most appropriate repair technique. This approach surpasses previous studies that focus solely on severity assessment, providing a more comprehensive framework for road maintenance management.

From a practical point of view, the implementation of this architecture has the potential to significantly improve road safety. It enables early detection of potholes, facilitates the planning of preventive repairs, and reduces long-term maintenance costs. In addition, institutions responsible for road maintenance management will benefit from reduced inspection times and the associated costs of early-stage damage repair.

For future research, we propose integrating positioning system (GPS) data to map and quantify potholes in specific areas. This data can be readily integrated into existing GIS platforms through a standardized export format, allowing pothole information (location, severity, and recommended repairs) to be visualized on interactive maps. This integration enables maintenance teams to view deterioration patterns across entire road networks and supports spatial analysis for maintenance planning. This approach would optimize resource usage and improve road infrastructure's durability and safety by comprehensively addressing damage at the roadway section level rather than focusing solely on individual deterioration.

Additionally, incorporating additional contextual data sources could improve the system's decision-making capability. Traffic density information would help prioritize repairs in high-traffic areas, where deterioration tends to accelerate. Weather data could contribute to predictive maintenance by identifying areas most susceptible to pothole formation. Integrating multiple data sources would enable a more comprehensive approach to maintenance planning.

This architecture has the potential to be implemented in governmental inspection programs focused on monitoring urban routes. In this context, vehicles could be equipped with depth cameras and advanced computational systems, such as the Nvidia Jetson Xavier, which is capable of processing data in real-time to optimize the detection and assessment of infrastructure damage. The information collected during the monitoring routes can be stored locally or transmitted in real-time to a central office, enabling engineers to analyze the results and make informed decisions for planning and executing corrective actions.

Such systems could be deployed by autonomous vehicles equipped with depth sensors for consistent roadside data collection or by drones with tailored parameters for aerial surveillance. Both implementations would require automated processing channels for real-time analysis and centralized data storage.

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Mario Roman-Garay: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Hector Rodriguez-Rangel:** Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Carlos Beltran Hernandez-Beltran:** Software, Methodology, Data curation. **Peter Lepej:** Writing – review & editing, Visualization, Methodology. **José Eleazar Arreyguez-Rocha:** Writing – review & editing, Visualization, Investigation. **Luis Alberto Morales Rosales:** Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Transportation Officials (1993). AASHTO Guide for Design of Pavement Structures, 1993, (Vol. 1). Aashto.
- [2] Y.J. Cha, R. Ali, J. Lewis, O. Büyükoztürk, Deep learning-based structural health monitoring, *Autom. Constr.* 161 (2024) 105328.
- [3] S.K. Ryu, T. Kim, Y.R. Kim, Image-based pothole detection system for ITS service and road management system, *Math. Probl. Eng.* 2015 (2015) 1–10.
- [4] G.B. Murali, V. SantoshKumar, D.N. Behera, K.K. Mohanta, O. Tulankar, S.S. Salimath, Pothole detection on roads using canny edge detection algorithm. *Applications of Computational Methods in Manufacturing and Product Design: Select Proceedings of IPDIMS 2020*, Springer Nature Singapore, Singapore, 2022, pp. 653–661.
- [5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gómez, ... and I. Polosukhin (2017) Attention is all you need. *Advances in neural information processing systems*, 30..
- [6] Y.J. Cha, W. Choi, O. Büyükoztürk, Deep learning-based crack damage detection using convolutional neural networks, *Comput. Aided Civ. Infrastruct. Eng.* 32 (5) (2017) 361–378.
- [7] Y.J. Cha, W. Choi, G. Suh, S. Mahmoudkhani, O. Büyükoztürk, Autonomous structural visual inspection using region-based deep learning for detecting multiple damage types, *Comput. Aided Civ. Infrastruct. Eng.* 33 (9) (2018) 731–747.
- [8] F.A.L.N. Mouzinho, and H. Fukai (2021). Hierarchical semantic segmentation based approach for road surface damages and markings detection on paved road. In: 2021 8th International Conference on Advanced Informatics: Concepts, Theory and Applications (ICAICTA) 1-5.IEEE..
- [9] R. Fan, H. Wang, M.J. Bocus, M. Liu, We learn better road pothole detection: from attention aggregation to adversarial domain adaptation. *Computer Vision-ECCV 2020 Workshops: Glasgow, UK, August 23-28, 2020, Proceedings, Part IV 16*, Springer International Publishing, 2020, pp. 285–300.
- [10] V. Pereira, S. Tamura, S. Hayamizu, and H. Fukai (2019). Semantic segmentation of paved road and pothole image using u-net architecture. In: 2019 International Conference of Advanced Informatics: Concepts, Theory and Applications (ICAICTA) 1-4.IEEE..
- [11] A. Mednis, G. Strazdins, R. Zviedris, G. Kanonirs, L. Selavo, Real time pothole detection using android smartphones with accelerometers. 2011 International conference on distributed computing in sensor systems and workshops (DCOSS), 2011, pp. 1–6 (IEEE).
- [12] C.H. Ho, M. Snyder, D. Zhang, Application of vehicle-based sensing technology in monitoring vibration response of pavement conditions, *J. Transp. Eng. Part B Pavements* 146 (3) (2020) 04020053.
- [13] R. Ali, Q. Bin-Saeed, O. Buyukozturk, S. Lee, and Y. Cha (2024). Monocular Computer Vision-Based Simultaneous Pothole Segmentation and 3d Volume Prediction Using 3dpredictnet. Available at SSRN 5045587..
- [14] S. Debnath, M. Paul, T. Debnath, Applications of LiDAR in agriculture and future research directions, *J. Imaging* 9 (3) (2023) 57.
- [15] X. Liang, V. Kankare, J. Hyyppä, Y. Wang, A. Kukko, H. Haggrén, M. Vastaranta, Terrestrial laser scanning in forest inventories, *ISPRS J. Photogramm. Remote Sens.* 115 (2016) 63–77.
- [16] M. Gehrig, W. Aarents, D. Gehrig, D. Scaramuzza, Dsec: a stereo event camera dataset for driving scenarios, *IEEE Robot. Autom. Lett.* 6 (3) (2021) 4947–4954.
- [17] G. Bosurgi, M. Modica, O. Pellegrino, G. Sollazzo, An automatic pothole detection algorithm using pavement 3D data, *Int. J. Pavement Eng.* 24 (2) (2023) 2057978.
- [18] H. Wu, L. Yao, Z. Xu, Y. Li, X. Ao, Q. Chen, B. Meng, Road pothole extraction and safety evaluation by integration of point cloud and images derived from mobile mapping sensors, *Adv. Eng. Inform.* 42 (2021) 100936.
- [19] J. Wang, S. Wang, K. Ma, Z. Wang, Perceptual depth quality in distorted stereoscopic images, *IEEE Trans. Image Process.* 26 (3) (2016) 1202–1215.
- [20] A. Georgopoulos, C. Ioannidis, A. Valanis, Assessing the performance of a structured light scanner. *International archives of photogrammetry, remote sensing and spatial, Inf. Sci.* 38 (Part 5) (2010) 251–255.
- [21] Y.M. Kim, Y.G. Kim, S.Y. Son, S.Y. Lim, B.Y. Choi, D.H. Choi, Review of recent automated pothole-detection methods, *Appl. Sci.* 12 (11) (2022) 5320.
- [22] F. Wan, C. Sun, H. He, G. Lei, L. Xu, T. Xiao, YOLO-LRDD: A lightweight method for road damage detection based on improved YOLOv5s, *EURASIP J. Adv. Signal Process.* 2022 (1) (2022) 98.
- [23] M. Ren, X. Zhang, X. Chen, B. Zhou, Z. Feng, YOLOv5s-M: a deep learning network model for road pavement damage detection from urban street-view imagery, *Int. J. Appl. Earth Obs. Geoinf.* 120 (2023) 103335.
- [24] G.H. Beckman, D. Polyzois, Y.J. Cha, Deep learning-based automatic volumetric damage quantification using depth camera, *Autom. Constr.* 99 (2019) 114–124.
- [25] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J.M. Alvarez, P. Luo, SegFormer: simple and efficient design for semantic segmentation with transformers, *Adv. Neural Inf. Process. Syst.* 34 (2021) 12077–12090.
- [26] 2023Libro: CSV. CONSERVACIÓN. Capítulo: 008. Determinación de los Deterioros Superficiales de los Pavimentos (DET). (M · CSV · CAR · 1 · 03 · 008/23). Secretaría de Infraestructura, Comunicaciones y Transporte. (.
- [27] A. Ahmed, M. Ashfaq, M.U. Ulhaq, S. Mathavan, K. Kamal, M. Rahman, Pothole 3D reconstruction with a novel imaging system and structure from motion techniques, *IEEE Trans. Intell. Transp. Syst.* 23 (5) (2021) 4685–4694.
- [28] J.E. Peralta-López, J.A. Morales-Viscaya, D. Lázaro-Mata, M.J. Villaseñor-Aguilar, J. Prado-Olivarez, F.J. Pérez-Pinal, A.I. Barranco-Gutiérrez, Speed bump and pothole detection using deep neural network with images captured through zed camera, *Appl. Sci.* 13 (14) (2023) 8349.
- [29] I. Katsamenis, A. Sakelliou, N. Bakalos, E. Protopapadakis, C. Klaridopoulos, N. Frangakis, ... and D. Kalogeras (2023). Deep transformer networks for precise pothole segmentation tasks. In: *Proceedings of the 16th International Conference on Pervasive Technologies Related to Assistive Environments* 596–602..
- [30] M.R. De Blasii, A. Di Benedetto, M. Fiani, Mobile laser scanning data for the evaluation of pavement surface distress, *Remote Sens.* 12 (6) (2020) 942.
- [31] F. Ozoglu, T. Gökgöz, Detection of road potholes by applying convolutional neural network method based on road vibration data, *Sensors* 23 (22) (2023) 9023.
- [32] C. Wu, Z. Wang, S. Hu, J. Lepine, X. Na, D. Ainalis, M. Stettler, An automated machine-learning approach for road pothole detection using smartphone sensor data, *Sensors* 20 (19) (2020) 5564.
- [33] Bhatt, U., Mani, S., Xi, E., and Kolter, J.Z. (2017). Intelligent pothole detection and road condition assessment. arXiv preprint arXiv:1710.02595.
- [34] J. Ha, D. Kim, M. Kim, Assessing severity of road cracks using deep learning-based segmentation and detection, *J. Supercomput.* 78 (16) (2022) 17721–17735.
- [35] J.S. Miller, and W.Y. Belling, (2003). Distress identification manual for the long-term pavement performance program (No. FHWA-RD-03-031). United States Federal Highway Administration. Office of Infrastructure Research and Development..
- [36] B. Dwyer, J. Nelson, (2022), J. Solawetz, , et al. Roboflow (Version 1.0) [Software] Available from (https://roboflow.com).computer vision..
- [37] M.A. Fischler, R.C. Bolles, Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography, *Commun. ACM* 24 (6) (1981) 381–395.
- [38] Mateusz Lango, Tackling the problem of class imbalance in multi-class sentiment classification: an experimental study, *Found. Comput. Decis. Sci.* 44 (2) (2019) 151–178, <https://doi.org/10.2478/fcds-2019-0009>.
- [39] Chih-Fong Tsai, et al., Under-sampling class imbalanced datasets by combining clustering analysis and instance selection, *Inf. Sci.* 477 (2019) 47–54, <https://doi.org/10.1016/j.ins.2018.10.029>.